Household Sorting, Mobility Costs, and Residential Misallocation*

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Abstract

This paper studies how moving costs affect household mobility, welfare, and residential sorting within U.S. metropolitan areas. We develop a dynamic model of neighborhood choice and estimate moving costs using data on more than 3.5 million homeowners from the American Community Survey. Our results show that realtor commissions substantially increase misallocation by reducing mobility, especially in high-cost housing markets. Counterfactual simulations indicate that halving commission rates would significantly increase mobility and generate large welfare gains. An analysis of optimal policy indicates that a simple, nationally set commission level could capture the majority of hypothetical first-best welfare gains.

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1 Introduction

Housing markets allocate economic opportunity. A household's residential location determines its access to jobs, schools, environmental quality, and other local public goods and amenities that affect the production and productivity of human capital (Chetty et al., 2016; Bayer et al., 2008; Finkelstein et al., 2021). For households that can afford to buy houses, ownership also provides an opportunity to build wealth. This opportunity is unevenly distributed, however, because housing appreciation rates vary greatly across neighborhoods within metropolitan areas (Landvoigt et al., 2015). For these reasons, household welfare depends crucially on how homeowners sort themselves over neighborhoods.

Given the importance of residential sorting for household welfare, it is striking, and perhaps concerning, that household mobility has declined dramatically in the United States. The annual migration rate has declined steadily for more than four decades, with a cumulative reduction of more than 50% between the early 1980s and the early 2020s. Most of the decline is due to fewer moves occurring within metropolitan areas. In principle, a decline in this local housing market "churn" could be welfare enhancing if it means that households are being better matched to neighborhoods, for example, due to the rise of online information (e.g. Zillow.com). On the other hand, the decline in churn could be welfare reducing if it means that rising moving costs are preventing households from adjusting to lifecycle shocks—such as marriage, childbirth and retirement—that change their preferences over neighborhoods.

One potential explanation for the decline in churn is that moving costs have risen substantially since 1980 due to realtor commissions. In the United States these commissions are typically fixed near 6% of a house's sale price (a high percentage by international standards). Because commissions scale proportionally with house values, rising home prices have mechanically increased the cost of moving, especially in high-cost metropolitan areas. The average commission approximately doubled in real terms between the early 1980s and the early 2020s. In 2024, for example, the

average home seller paid $32,000 in realtor commissions, similar to what the average household pays for a car.

The purpose of this paper is to examine how realtor commissions affect household mobility, welfare, and the allocation of households to higher- and lower-amenity neighborhoods. We develop a dynamic model of residential sorting that is designed to identify the utility cost of moving locally under minimal assumptions about the structure of preferences, beliefs, and the evolution of household demographics. We use the model to provide the first national estimates for moving costs among homeowners in U.S. metropolitan areas. Then we design counterfactual experiments to investigate the implications of alternative realtor commission structures.

Despite longstanding interest in migration frictions, surprisingly little is known about the magnitude and composition of moving costs within U.S. metropolitan areas. Prior studies that estimated the utility cost of moving focused almost exclusively on migration between metropolitan areas (e.g. Bishop, 2013; Bayer et al., 2009; Kennan and Walker, 2011; Liang et al., 2024). While those long-distance moves are important for labor market adjustment, they account for a very small fraction of domestic migration. Moreover, the utility cost of moving between metropolitan areas may be much higher than the cost of moving locally due to wage differentials, physical moving costs, and the psychological cost of moving away from family, friends, and a familiar area. The few existing studies that estimated local moving costs focused on narrow subpopulations and specific geographic areas such as low-income renters in Boston and Seattle (e.g. Galiani et al., 2015; Bergman et al., 2024) or White medium-income homeowners in the San Francisco Bay Area (Bayer et al., 2016).

Our analysis uses data from more than 3.5 million homeowners in the American Community Survey (ACS) between 2013 and 2018. We focus primarily on within-metropolitan moves across Public Use Microdata Areas (PUMAs) in 150 metropolitan housing markets. Our estimation sample covers roughly two-thirds of U.S. homeowner households. We combine household demographics, migration

histories, and neighborhood characteristics from the ACS with additional PUMA-specific measures for the local user cost of housing (Bishop et al., 2025), exposure to fine particulate air pollution (Di et al., 2017), and Opportunity Atlas measures of children's long-run socioeconomic outcomes (Chetty and Hendren, 2018).

We start by documenting four facts about realtor fees and homeowner mobility. First, realtor fees are a surprisingly large share of housing expenditures. They account for about 15% of the average annualized cost of homeownership. Second, the spatial variation in fees is tremendous. While the services that realtors provide are fairly homogeneous, their typical compensation varies by more than an order of magnitude across PUMAs, from roughly $11,000 in the bottom quartile of PUMA housing value to over $100,000 in the top quartile. Third, realtor fees doubled over the last 40 years while the homeowner mobility rate halved. Fourth, mobility rates are negatively associated with realtor fees, both within and across housing markets. Taken together, these facts suggest that realtor fees might have important effects on homeowner churn and household welfare. Measuring these effects requires a model.

We develop a dynamic model of residential sorting where, each period, a homeowner must choose whether to remain in their current house, move to a new house in the same neighborhood, or move to a different neighborhood in the same housing market. The homeowner faces a utility cost of moving within their market that includes the need to pay 6% of the sale price of their house to real estate agents. Homeowners may be forward looking over the evolution of neighborhood amenities and housing prices. We allow for flexible heterogeneity in neighborhood preferences, moving costs, and expectations.

Our estimator is designed to overcome two main challenges in identifying structural parameters describing sorting behavior. First, housing prices are endogenous because they capitalize unobserved neighborhood quality. Second, since realtor fees are proportional to housing prices, they are mechanically collinear with the flow cost of housing consumption. We address these challenges in two ways. First, we use

data on spatial variation in local user cost rates from Bishop et al. (2025) to break the collinearity between housing expenditures and realtor commissions. Intuitively, two neighborhoods with identical housing prices and realtor fees can differ in the annualized cost of homeownership due to differences in their local property tax rates and housing appreciation rates. Second, we leverage econometric insights from Hotz and Miller (1993) and Bayer et al. (2016) to separate the neighborhood choice problem from the move-stay decision. Intuitively, movers' destination choices flexibly identify the flow utility of each neighborhood, while the overall propensity to move identifies mobility costs. This strategy effectively differences out unobserved neighborhood quality from the moving decision and recovers moving costs from revealed preferences over relocation behavior.

Our estimates imply that realtor commissions substantially limit housing market churn. Conditional on the commission size, we also find substantial demographic heterogeneity in moving costs. For example, we find that moving costs are higher for older households, married households, and households with school-aged children, consistent with stronger attachments to existing social networks, schools, and local amenities. Higher-income households exhibit lower baseline moving costs but are also more sensitive to realtor commissions because they tend to reside in more expensive neighborhoods. These frictions help explain the lower rates of mobility in expensive neighborhoods and metropolitan areas, and imply that rising house prices have amplified residential misallocation through the realtor commission system.

We use the estimated model to study counterfactual changes to realtor commission structure. First, we investigate the effects of reducing the fee from 6% to 3%, closer to international norms. We find that halving the fee would increase within-market churn by approximately 32% on average, with especially large effects in high-price markets such as Los Angeles and New York. Reduced moving costs generate substantial welfare gains, increasing aggregate consumer surplus by roughly $40.7 billion per

year.[1] The gains are largest in high-cost metropolitan areas that experience the largest nominal reductions in realtor fees. We then examine how halving the fee would change aggregate commissions. Although lower fees increase transaction volume through greater mobility, aggregate commissions decline by approximately 28%, or roughly $18.7 billion annually. However, the effects are highly heterogeneous. In high-price metro areas, increased transaction volume substantially offsets the decline in commission per sale. In a few metro areas, this even leads to an increase in aggregate commissions. We then show that instituting a flat fee equivalent to the current median realtor fee would have similar effects on total welfare, but with smaller losses for realtors. We also show that the increase in commissions since 1986 that was caused by the appreciation of housing values can explain roughly 14% of the decline in US migration since 1986.

We also consider the idea of politically feasible optimal policy. That is, we consider how to design an optimal realtor compensation structure under the constraint that the policy cannot reduce total realtor commissions from their current level. As a benchmark, we consider the unrealistic, first-best scenario where a different fee can be charged in every neighborhood PUMA in our sample and choose those approximately 1,800 fees to maximize total welfare subject to the policy not reducing total realtor commissions. We then solve for the optimal single national flat fee that would maximize total welfare subject to the constraint. Strikingly, we find that a flat fee of $24,100 would capture 98% of these gains. Further constraining the policy to ensure household welfare increases in every neighborhood only reduces the welfare gains to 76% of the first-best outcome.

Our study contributes to three distinct literatures. First, it contributes to the literature on residential sorting and amenity demand. A large literature studies how households sort across neighborhoods in response to local amenities and housing

[1]All dollar amounts in the paper are expressed in constant 2026 dollars using the Consumer Price Index for All Urban Consumers (CPI-U). Unless noted otherwise, aggregate dollar figures are annual totals for the population of our 150-market study area, computed with ACS household weights.

costs (Kuminoff et al., 2013). Our study is most closely related to a small number of studies that model within-city sorting dynamics (Mastromonaco, 2014; Bayer et al., 2016; Caetano, 2019; Davis et al., 2021; Almagro et al., 2022; Almagro and Domínguez-Iino, 2025). Our econometric approach is simpler than prior dynamic studies in the sense that we avoid the need to decompose flow utility into marginal values for individual amenities. We also extend the residential sorting literature by adapting the dynamic discrete-choice estimator from Bayer et al. (2016) to identify the utility cost of moving from public-use microdata on households, as opposed to the proprietary longitudinal data used in prior studies. In addition, our counterfactuals connect mobility frictions to changes in amenity exposure and spatial inequality at the national level. To do this, we go beyond Bayer et al. (2016) and develop a framework to allow for equilibrium counterfactual simulations.

Second, our paper contributes to the literature on migration frictions and mobility policies (e.g. Bishop, 2013; Bayer et al., 2009; Kennan and Walker, 2011; Galiani et al., 2015; Liang et al., 2024; Bergman et al., 2024). As we noted earlier, prior studies focused on migration between metropolitan areas, and migration by specific demographic groups within a couple of specific metropolitan areas. We add to this literature by providing the first national estimates of within-metropolitan moving costs for homeowners and quantifying the contribution of realtor commissions to residential mobility.

Finally, our paper contributes to the literature on realtor behavior and housing market institutions. Prior work examined the structure of brokerage markets, agent incentives, and the persistence of uniform commission rates rather than the allocative consequences of commissions for household sorting and welfare (Hsieh and Moretti, 2003; Levitt and Syverson, 2008; Jia and Pathak, 2010; Barwick and Pathak, 2015; Barwick et al., 2017; Kim, 2025; Buchak et al., 2024). In contrast, we study the equilibrium consequences of realtor commissions for residential allocation and welfare. Our estimates imply that commissions are not merely transfers between households

and intermediaries, but instead constitute quantitatively important distortions that reduce mobility, impede efficient sorting, and generate substantial deadweight losses.

2 Background, Data, and Descriptive Facts

2.1 Background

Moving costs, capturing both the financial and psychological costs or moving, are typically estimated using national sorting models such as Bayer et al. (2009), Kennan and Walker (2011), and Bishop (2013). They develop models where households maximize utility by choosing residential location. If a chosen location is different from the household's current location, then it must pay a utility cost of moving. In Bayer et al. (2009), Kennan and Walker (2011), and Bishop (2013) estimates for moving costs are in the hundreds of thousands of dollars. Intuitively, this is the amount households would pay to avoid being reassigned to a randomly-chosen city somewhere in the US. It conflates the cost of hiring a moving company, payments to real estate agents and bank brokers, potentially lower wages from changing jobs, and the psychic cost of moving away from family, friends, and a familiar environment. In this context, large costs make sense.

In comparison, little is known about the costs of moving *within* metro areas. This is an important knowledge gap because the vast majority of moves are local. For example, approximately 80% of all moves are contained within a single metropolitan housing market.[2] Since housing market churn accounts for the vast majority of internal US migration, it is important to understand this phenomenon.

We aim to estimate the cost of moving within metro areas and to isolate the contribution of realtor fees. Therefore, some background on realtor fee institutions

[2]Specifically, this is the share of moves that occur within Metropolitan Statistical Areas (MSAs) or Combined Statistical Areas that aggregate adjacent MSAs that have integrated housing and labor markets. Figure A.1 shows that the 80% statistic has been stable over time and applies to both moves and realtor commissions.

may be helpful. In the US about 90% of buyers and sellers use real estate agents. The total fee paid is typically 6%, which is large by international standards, and hardly varies within the US. The 6% convention has been in place since the 1950s following a recommendation by the National Association of Realtors (NAR). In practice, the 6% fee is typically split evenly between the buyer's agent and the seller's agent. But it is paid entirely by the seller; the buyer does not pay anything directly. The seller pays 6% to their agent, who then gives half to the buyer's agent. Typically, the seller's agent helps with pricing strategy, marketing the house with photos and staging it for visitors, managing showings, communicating offers to sellers, and giving advice on negotiation. The buyer's agent traditionally helps them search over neighborhoods and houses, arranges for them to tour houses, gives advice on negotiation, and general advice like who to call for inspection and financing.

Traditionally, buyers did not have the ability to negotiate fees down. This led the US Department of Justice to bring an antitrust case against NAR during the Biden administration. NAR agreed to pay $418 million and to give buyers the ability to negotiate. However, in practice, this seems to have had a very limited effect so far and 6% remains widely used (Buchak et al., 2024; Kim, 2025; Duarte and Zhang, 2025; Smith and Jain, 2025). Going forward, there may be future entry of low-cost realtor technologies or, if not, future federal administrations may take a more forceful approach to restructuring the industry.

2.2 Data

We combine data from several sources. This section briefly summarizes how we constructed the estimation sample. Appendix A provides additional details.

2.2.1 Sample Construction and Market Definition

We start with micro data from the annual American Community Surveys (ACS) from 2013-2018. For each homeowner, we see their current location at the level of a

public use microdata area (or PUMA) which is a spatially contiguous area containing about 100,000 people. This will be our measure of residential neighborhood. We use ACS data on the household head's age, race, income, education, whether they have school-age children, whether they moved in the last year and, if they moved, where they moved from.

We then assign each household to a metropolitan housing market. This assignment starts by mapping each PUMA to a Metropolitan Statistical Area (MSA).[3] Then, in cases where the US Office of Management and Budget aggregates adjacent MSAs into a Combined Statistical Area (CSA) we define the CSA as the local housing market.

Finally, we make some data cuts to derive a national sample that is well suited to studying homeowner churn. Specifically, we focus on household heads who own their homes and live in multi-PUMA housing markets. We drop heads who move internationally, live outside metro areas, are under 18, have household incomes below zero or over $1 million, belong to rare race or ethnic categories, cannot be mapped to a single housing market, or who we observe moving between housing markets. The most important of these exclusions, by far, is our decision to drop the 25% of homeowner households that live in rural areas. While they are a non-trivial share of the homeowner population, they are less relevant for our study because, by definition, they live in thinly-populated areas with limited opportunities for local moves.

The resulting estimation sample contains more than 3.5 million homeowners living in 1,832 PUMAs and 150 housing markets. They represent 68% of all homeowners in the ACS. The average household lives in a market with 12 PUMAs, and the largest market—the New York / Newark CSA—contains 175 PUMAs.

[3] A small number of households residing in PUMAs that span multiple MSAs are excluded. See Table A.1 for details.

2.2.2 Neighborhood Housing Prices and Amenities

We measure the price of housing in each neighborhood using ACS data on mean house value. We also use ACS data to construct several measures of neighborhood demographic composition that household may perceive as amenities, or that may proxy for latent amenities (Bayer et al., 2007). These measures include mean income, the share of household heads with college degrees, share White, mean age, share over age 65, mean number of rooms, and median house age.

We collect additional data on PUMA-level amenities from three sources. First, as a measure of environmental quality, we use data on fine particulate air pollution ($PM_{2.5}$) from Di et al. (2017). $PM_{2.5}$ is commonly thought to be the most harmful type of air pollution for human health. It can penetrate deep into the lungs, and even into the brain. Next, as a measure of economic opportunity, we use several variables from the Opportunity Atlas developed in Chetty and Hendren (2018). These are causal measures of how living in a particular neighborhood as a child affects adult outcomes, conditional on parental income. We focus on four outcomes: incarceration, teen pregnancy, family income at age 29 and employment at age 29. Finally, as a control for latent amenities, we use the PUMA-specific user cost rate of housing from Bishop et al. (2025). The user cost rate depends on the cost of equity and financing a house via a mortgage, local property taxes, depreciation, a risk premium, expected appreciation, and tax subsidies to homeownership. Because many of these factors vary across space due, in part, to local amenities, so does the user cost rate, making it a useful all-purpose control for latent amenities.[4]

2.3 Descriptive Facts about Mobility and Realtor Fees

Four facts about homeowner mobility and realtor fees motivate our empirical approach. First, realtor fees account for a surprisingly large share of annualized housing costs—

[4]The user cost rate tends to be highest in the northeast and lower in the West and Pacific Coast. This can be seen in Appendix Figure A.2.

15%. We calculated the 15% statistic by taking the average realtor fee, amortizing it over the average duration a household lives in their house and comparing it to the annual real cost of homeownership calculated in Bishop et al. (2025) which accounts for the cost of land and structures as well as property taxes, financing costs, maintenance and depreciation. It is striking that people pay realtors so much when many of their traditional services are now provided for free by AI and websites like Zillow and Redfin.[5]

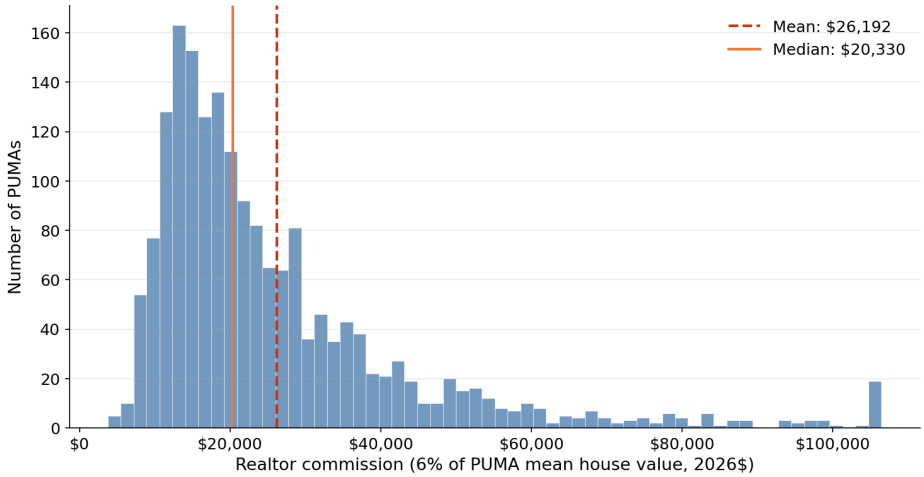
Our second fact is that, because the 6% fee is fixed, and housing prices vary greatly across space, the nominal realtor fee varies greatly across the US. Figure 1 shows the distribution of realtor fees across PUMAs in our estimation sample, calculated as 6% of average house value. In expensive neighborhoods of coastal housing markets like San Francisco, Los Angeles and New York, realtors make well over $100,000 per sale. In inexpensive neighborhoods of interior housing markets, realtors make well under $15,000 per sale. Yet they perform very similar tasks in both types of areas. This degree of spatial variation in the price of a service is rare and could induce substantial variation in household mobility.

Our third fact is that, over the last 40 years, the average realtor fee rose substantially in real terms, due to housing appreciation, while homeowner migration fell. Figure 2 shows this negative correlation. We aim to learn whether this correlation is causal and, if so, how much of the decline in mobility can be explained by the rise in realtor fees.

Finally, we use our estimation sample to document that, conditional on household type, mobility rates are higher in areas where realtor fees are lower. This can be seen in Figure 3. The fitted regression lines show that higher prices, and therefore higher realtor fees, are associated with lower mobility. This association holds both within and between housing markets.

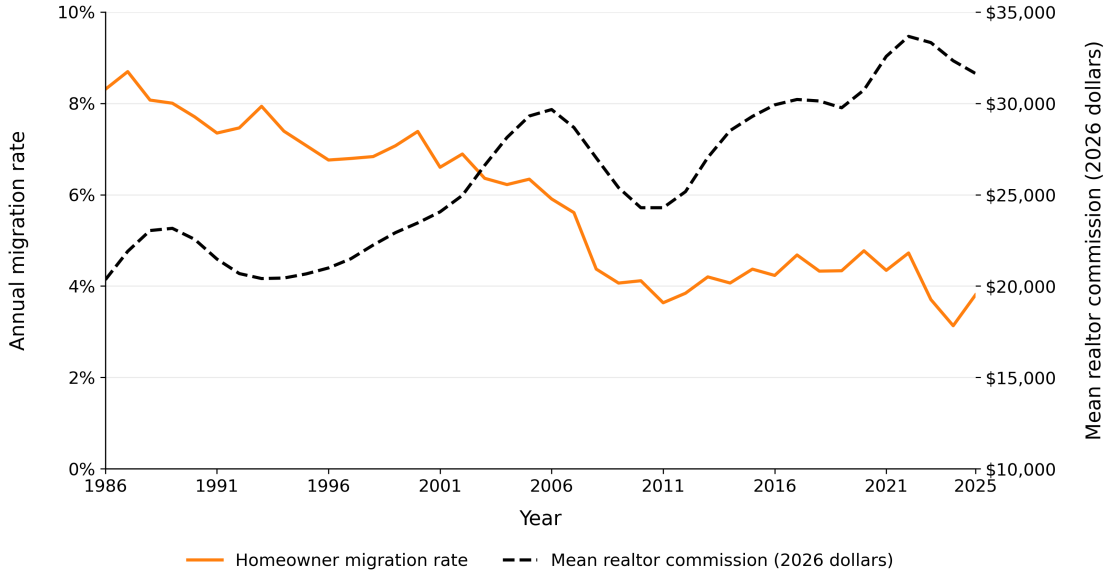
[5]Zillow and Redfin give buyers detailed information about houses and neighborhoods, buyers and sellers can see how much comparable houses sell for, and AI can help with negotiation and strategy.

Figure 1: Distribution of PUMA-Level Realtor Fees



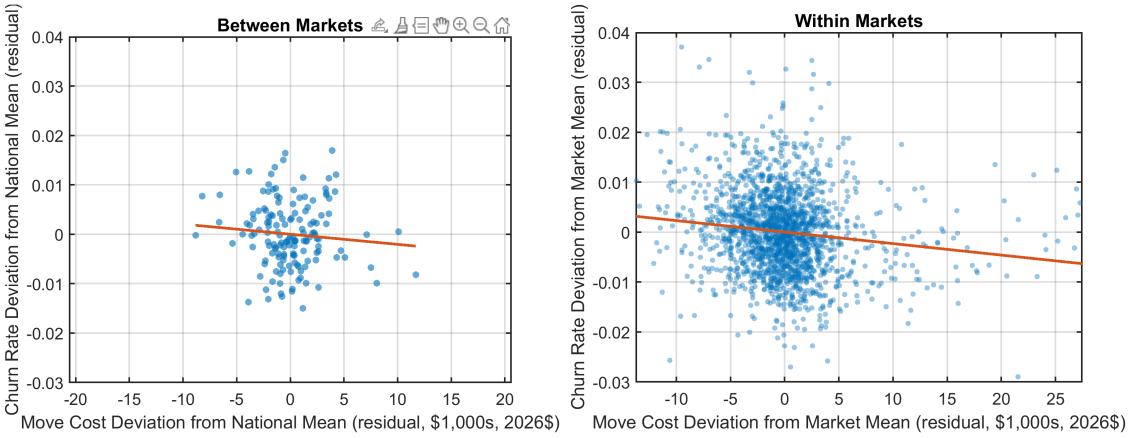
Notes: Each observation is a PUMA. The fee is 6% of the PUMA's mean owner-occupied house value in the pooled 2013–2018 ACS data, expressed in constant 2026 dollars, and the sample is restricted to the 150 metropolitan markets used to estimate the model. For scale, the 19 PUMAs (1%) with fees above the 99th percentile ($106,400) are grouped into the rightmost bin; no PUMAs are dropped, and the maximum fee is $174,200.

Figure 2: Homeowner Migration Rate and Average Realtor Commission



Notes: The solid line shows the share of homeowner household heads who moved in the past 12 months, calculated from Current Population Survey micro data from 1986 to 2024. The year 1995 is dropped because the Census Bureau flags its migration data as suspect. The dashed line shows 6% of the average sales price of houses sold in the US (FRED series ASPUS), deflated to 2026 dollars with the CPI-U and smoothed with a 3-year centered moving average. The figure is based on the entire US population of homeowners, rather than our estimation sample, because the CPS data are too geographically coarse to map into our definition of markets.

Figure 3: Churn Versus Moving Cost, Between and Within Markets



Notes: Both panels partial out the following household demographics via OLS regression on the estimation sample: PUMA-level means of household income, the share of household heads who are seniors, the share married, the share with school-age children, and race/ethnicity shares (Black, Asian, Hispanic; White non-Hispanic). The two panels share the same vertical axis and the same horizontal-axis width, so the slopes of the fitted lines are directly comparable. While all estimation sample PUMAs are used in the regression, for scale, the right panel does not display 13 of the 1,832 PUMAs (9 with move-cost deviations above $27,400 and 4 below $-\$13,700$). Move costs are in constant 2026 dollars.

The left panel of Figure 3 is a between-market analysis that compares a market's mobility rate with the market's mean realtor fee. The right panel is a within-market analysis that compares a PUMA's mobility rate with a PUMA's mean realtor fee, while controlling for market fixed effects. Both panels control for household demographics that could affect mobility such as income, age, race/ethnicity, marital status, and presence of school-age children. The figure shows that, between markets, a $10,000 increase in the mean realtor fee is associated with a 0.21 percentage point reduction in mobility. Within markets, the association is 0.23 percentage points, which is roughly 5% of the mean within-market mobility rate of 4.7%.

Taken together, these four descriptive facts suggest that realtor fees might have important effects on homeowner churn and household welfare. Measuring these effects requires a model.

3 Model

We model forward-looking households choosing neighborhoods in the presence of moving frictions. Before formally presenting the model and estimator, we first summarize the identification challenges and preview our research design.

3.1 Identification Challenge

Estimating how moving costs affect mobility is difficult because neighborhood attributes (both observed and unobserved) may simultaneously determine the utility of living in a neighborhood and the disutility of moving out. This identification challenge is particularly problematic when identifying the impact of realtor fees as housing prices simultaneously affect both the value of living in a location and the cost of moving out of it.

To illustrate the difficulty of separately identifying the distinct effects of housing and neighborhood attributes on flow utility and moving costs, consider the following regression equation: The dependent variable, $move_{ij}$ is an indicator denoting whether household i moves out from neighborhood j. For simplicity, make the temporary assumption that we know what is the best alternative neighborhood and denote it with k. Then, the regression of moving out of j is given by:

$$move_{ij} = \underbrace{v_i(X_k, \xi_k, ucr \cdot P_k)}_{\text{value of moving to } k} - \underbrace{v_i(X_j, \xi_j, ucr \cdot P_j)}_{\text{value of staying in } j} - \underbrace{MC_i(rfr \cdot P_j)}_{\text{moving costs from } j} + \epsilon_{ijk} \tag{1}$$

where X_j denotes observable neighborhood characteristics, ξ_j denotes unobservable neighborhood attributes, P_j is housing value, ucr is the user-cost rate, and rfr is the realtor fee rate. The user-cost rate is the scalar that converts house value into an annualized cost (Poterba, 1984; Bishop et al., 2025).

Two econometric problems can be seen in Equation (1). First, if ucr and rfr do not vary at the neighborhood level, then the rank condition fails because realtor fees,

$rfr \cdot P_j$, are collinear with annualized housing costs, $ucr \cdot P_j$. In other words, P_j plays two roles: it increases the annualized cost of living in j making it less attractive to stay but also increases the realtor fee making it more attractive to stay. The second challenge is that the order condition fails because housing values, P_j are correlated with unobserved neighborhood quality, ξ_j. High-priced neighborhoods tend to possess desirable amenities that are difficult to observe directly; these unobserved amenities are capitalized into house values and also directly affect the probability of moving.

3.2 Research Design Overview

Without additional data, one cannot separately identify the value households derive from neighborhoods and the costs they incur when moving using a simple mobility regression such as Equation (1). A key contribution of this study is a new identification strategy that overcomes these challenges. First, we add data on spatially varying user costs rates, ucr_j, from Bishop et al. (2025). Second, and more importantly, we use information on the location chosen by a household conditional on moving (Galiani et al., 2015).

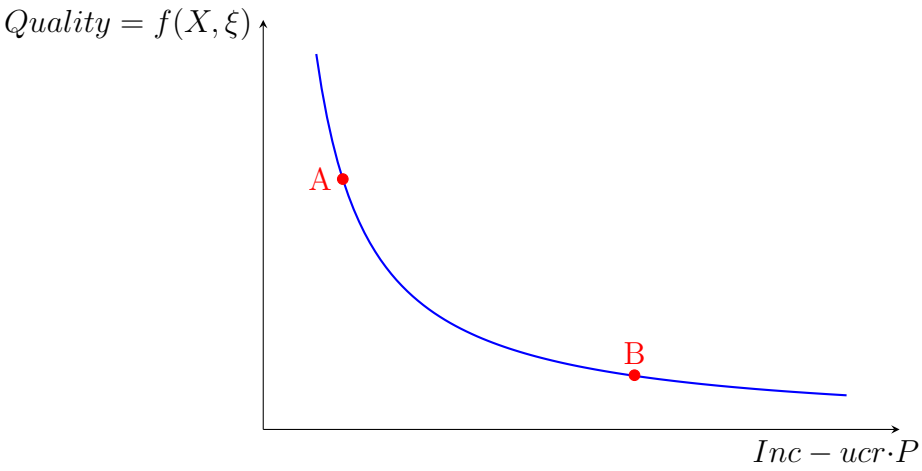
The approach builds on the frameworks of Hotz and Miller (1993), Bayer et al. (2016), and Schiraldi (2011). The high-level overview is that (i) neighborhood choice, conditional on moving, identifies neighborhood preferences and (ii) once the value of living in a neighborhood is identified from location choices, variation in moving identifies mobility costs separately.

The variation that identifies moving costs is therefore variation in $rfr_j \cdot P_j$ while holding utility constant. The intuition behind why it is valid to vary prices while holding utility constant can be seen in Figure (4) which illustrates an indirect indifference curve, $v[f(X, \xi), Inc - ucr \cdot P] = \delta$. On this indirect indifference curve, the vertical axis is neighborhood quality (depending on X and ξ) and the horizontal axis is numeraire consumption, i.e., income less housing expenditures. Consider two types of neighborhoods, A and B, where A neighborhoods have better amenities.

Since A neighborhoods have better amenities than B neighborhoods, they will have higher prices in equilibrium. So A neighborhoods are high-price, high-amenity neighborhoods and B neighborhoods are low-price, low-amenity neighborhoods. As they lie on the same indifference curve, neighborhoods give the same utility. However, when we add time varying household shocks, mobility out of A neighborhoods will be lower because homeowners there have to pay higher realtor fees. Therefore, holding utility constant, there will be less churn in A neighborhoods compared with B neighborhoods.

Another way to see this is that if there are zero moving costs, then the stock share and flow share for each neighborhood will be equal. If moving costs are present, then stock shares will be higher (lower) than flow shares in A (B) neighborhoods. The correlation between the magnitude of this difference and $rfr \cdot P$ identifies how realtor fees determine moving costs.

Figure 4: Indifference Curve for $v[f(X,\xi), Inc - ucr \cdot P] = \delta$



3.3 Model Components

Our model is one of within-market migration within the 150 metropolitan housing markets described in Section 2. Households have observable attributes, Z_i, and maximize lifetime utility by making location decisions over neighborhoods, (PUMAs).

Households can stay in their existing location, move within neighborhood, or move to a new neighborhood. Households that move incur mobility costs, both psychological and financial. As discussed above, neighborhoods are described by attributes X_j, ucr_j, P_j and ξ_j.

Let the lifetime utility that household i, with observable attributes, Z_i receives from living in neighborhood j when paying no moving costs be given by:

$$\tilde{V}_{ij} = v_{ij} + \epsilon_{ij} \tag{2}$$

v_{ij} flexibly captures observable and unobservable attributes of neighborhoods (such as X_j, ucr_j, P_j, and ξ_j) and observable attributes of households (such as Z_i). Household-specific unobserved heterogeneity is captured by ϵ_{ij}.

h_i denotes a household's prior location. Moving from neighborhood h_i to neighborhood j reduces utility by an amount MC_{ih_i}. Moving costs can vary with Z_i, by origin neighborhood, and have a stochastic component. Although we allow moving costs to vary with Z_i, incurring moving costs does not causally affect Z_i and we assume that MC_{ih_i} is not a function of the destination neighborhood, j.[6]

Households may choose to move within their own neighborhood. Therefore, if there are J neighborhoods, there are $J+1$ choice alternatives and $J+1$ choice shocks in ϵ_i. Following Bayer et al. (2016), we simplify notation and do not use a separate index for neighborhood and choice. The choice of not moving is denoted by $J+1$. The choice of j for $j < J+1$ denotes moving and choosing neighborhood j.

The full lifetime utility that household i, which begins a period in neighborhood h_i, receives from choosing alternative/neighborhood j is then given by:

$$V_{ij} = v_{ij} - MC_{ih_i} 1_{[j \neq J+1]} + \epsilon_{ij} \tag{3}$$

[6]The assumption that MC does not vary with destination neighborhood is convenient, but not necessary.

We assume that ϵ_{ij} is distributed i.i.d., Type 1 Extreme Value and that moving costs are additively separable in an unobservable term, η_i, such that $MC_{ih_i} = MC_{ih_i} + \eta_i$. Therefore, the probability that household i chooses neighborhood j, when beginning the period in neighborhood h_i, is given by:

$$P_{ij} = \int \frac{exp(v_{ij} - (MC_{ih_i} + \eta_i)1_{[j \neq J+1]})}{exp(v_{ih_i}) + \sum_{k \neq J+1} exp(v_{ik} - (MC_{ih_i} + \eta_i))} d\eta_i \tag{4}$$

and the probability of moving, when beginning the period in neighborhood h_i, is given by:

$$P_i^{move} = \int \frac{\sum_{k \neq J+1} exp(v_{ik} - (MC_{ih_i} + \eta_i))}{exp(v_{ih_i}) + \sum_{k \neq J+1} exp(v_{ik} - (MC_{ih_i} + \eta_i))} d\eta_i \tag{5}$$

Finally, as a preliminary specification, we parametrize moving costs as $MC_{ih_i} = \gamma_{0i} + \gamma_{1i} P_{h_i} rfr$. This allows there to be a household-specific baseline moving cost, γ_{0i} which varies with Z_i. In addition, Realtor fees, $P_{h_i} rfr$ also affect moving costs heterogeneously as governed by household-specific slope coefficient, γ_{1i}

3.4 Welfare

Our model can be used to calculate the welfare gains and losses of counterfactual policy scenarios. We consider indirect and direct measures of welfare. As an indirect measure of the welfare effects of an alternative policy, we can calculate the change in mobility rates caused by counterfactual policy. This measure complements the direct measures of welfare effects, i.e., changes to expected consumer and producer surplus. The goal of these welfare measures is to describe the level of misallocation under the current system and to describe how it could be improved under alternative mechanisms.

We consider various policies that change the realtor commission rate, rfr. These alternative policies affect behavior and utility through two mechanisms. First, rfr directly enters the utility function through moving costs MC_{ih_i}. Second, any change

to rfr will cause households to re-sort and so may affect both house values, P, as well as lifetime utilities, v.

3.4.1 Mobility Effects

Our first way to characterize effects on consumer welfare is to describe how much mobility would change from an alternative policy, rfr'. We can do this by plugging into Equation (5) our parameterization of moving costs.

$$P_i^{move}(rfr) = \frac{\sum_{k \neq J+1} exp(v_{ik}(rfr) - (\gamma_{0i} + \gamma_{1i} P_{h_i}(rfr) rfr)))}{exp(v_{h_i}(rfr) + \sum_{k \neq J+1} exp(v_{ik}(rfr) - (\gamma_{0i} + \gamma_{1i} P_{h_i}(rfr) rfr))} \quad (6)$$

The effect on total mobility can then be calculated as:

$$\Delta P^{move}(rfr', rfr) = \sum_i \left(P_i^{move}(rfr') - P_i^{move}(rfr) \right) \quad (7)$$

3.4.2 Expected Consumer Surplus

Beginning with Equations (2) and (3) and using the parameterization that $MC_{ih_i} = \gamma_{0i} + \gamma_{1,i} P_{h_i} rfr$, we can rewrite Equation (3) as

$$V_{ij} = v_{ij} - (\gamma_{0i} + \gamma_{1i} P_{h_i} rfr) 1_{[j \neq J+1]} + \epsilon_{ij} \quad (8)$$

The Emax of V_{ij} (ignoring the Euler–Mascheroni constant) is then given by:

$$E_\epsilon[max\{V_{ij}\}] = log\Big(\sum_k exp(v_{ik} - (\gamma_{0i} + \gamma_{1i} P_{h_i} rfr) 1_{[k \neq J+1]}) \Big) \quad (9)$$

which can be rewritten as:

$$E_\epsilon[max\{V_{ij}\}] = log\Big(exp(v_{h_i}) + \sum_{k \neq J+1} exp(v_k - (\gamma_{0i} + \gamma_{1i} P_{h_i} rfr)) \Big) \quad (10)$$

For a given policy, rfr, the expected consumer surplus for household i, measured

in dollars, is given by:

$$ECS_i(rfr) = (1/\gamma_{1i})log\Big(exp(v_{h_i}(rfr)) + \sum_{k \neq J+1} exp(v_k(rfr) - (\gamma_{0i} + \gamma_{1i} P_{h_i}(rfr) rfr))\Big) \tag{11}$$

Therefore, for any actual policy, rfr, we can calculate the total welfare impact to homeowners from an alternative policy, rfr', as:

$$\Delta ECS(rfr', rfr) = \sum_i \big(ECS_i(rfr') - ECS_i(rfr)\big) \tag{12}$$

Finally, we note that these calculations assume that reducing the Realtor rate, rfr, has no negative effect on household sorting. That is, any help that realtors provide with the matching process is invariant to changing rfr to rfr'.

3.5 Realtor Welfare

We also need to think about the effect on realtor welfare. We consider two potential scenarios which bound the effect on Realtor welfare.

The first potential scenario is that realtors face no barrier to entry and so the marginal entrant makes no excess profits. If one extends this to assume that the average Realtor makes no profit, then total Realtor welfare will be unchanged by any policy shift from rfr to rfr'.

The second scenario is that we calculate the Realtor welfare loss as the change in total expected commissions paid. Total expected commissions, for a given Realtor fee, rfr, are given by:

$$Comm(rfr) = \sum_i P_i^{move}(rfr) P_{h_i}(rfr) \tag{13}$$

and the effect of a policy shift from rfr to rfr' is given by:

$$\Delta Comm(rfr', rfr) = Comm(rfr') - Comm(rfr) \tag{14}$$

Assuming that scenario one (two) weakly underestimates (overestimates) the absolute value of welfare losses, the true effect is bounded by [0, $\Delta Comm(rfr', rfr)$]

4 Identification and Estimation

Building on the insights of Bayer et al. (2016), the model can be estimated in two stages. In Stage 1, we focus on movers for whom the probability of choosing j, given $j \neq J+1$ is given by:

$$P_{ij|d_i \neq J+1} = \int \frac{exp(v_{ij} - (MC_{ih_i} + \eta_i))}{\sum_{k \neq J+1} exp(v_{ik} - (MC_{h_i} + \eta_i))} d\eta_i = \frac{exp(v_{ij})}{\sum_{k \neq J+1} exp(v_{ik})} \tag{15}$$

Under the assumption that MC does not vary with destination location, the moving costs, $MC_{ih_i} + \eta_i$, do not affect the decision of where to locate, conditional on moving, and therefore drop out. Then, this is a simple conditional logit model where it is straightforward to use mobility outcomes to recover estimates of the v_{ij} terms.

We estimate $\hat{P}_{ij|d_i \neq J+1}$ using a conditional logit model where each household, conditional on being a mover, chooses between the J neighborhoods in their market. We estimate a flexible logit model where we include a neighborhood-specific fixed effect and a rich set of neighborhood-specific attributes interacted with our demographic household attributes.[7] In practice, we parameterize v_{ij} as:

$$v_{ij} = (Z_i' \Theta) X_j + \xi_j \tag{16}$$

[7]While it is a positive feature of the model that we flexibly include so many household attributes in the interactions with neighborhood-specific attributes, the curse of dimensionality preludes us from estimating a truly non-parametric model.

where Θ is a matrix of coefficients (of dimension equal to $dim(Z_i) \times dim(X_j)$) that governs how the interactions between Z_i and X_j affect neighborhood choice probabilities.

In Stage 2, we use the move/stay decision for all households to estimate moving costs, MC_{ih_i}. This mobility probability is given by:

$$P_i^{move} = \int \frac{\sum_{k \neq J+1} exp(v_{ik} - (MC_{ih_i} + \eta_i))}{exp(v_{ih_i}) + \sum_{k \neq J+1} exp(v_{ik} - (MC_{ih_i} + \eta_i))} d\eta_i \tag{17}$$

To complete the model, we need to make an assumption about the distribution of the moving cost shock, η_i. For example, if η_i is set to zero, this is a binary logit; if η_i is distributed a la Cardell (1997), this is a nested logit; if η_i is distributed normally, this is a mixed-logit.

As a preliminary specification, we parameterize moving costs as $MC_{ih_i} = \gamma_{0i} + \gamma_{1i} P_{h_i} rfr$. Estimation is straightforward and γ_i may be estimated using a binary logit model where the net utility of moving is:

$$log(\sum_{k \neq J+1} exp(v_k)) - v_{h_i} - \gamma_{0i} - \gamma_{1i} P_{h_i} rfr \tag{18}$$

and where the v_i terms are estimated in the first stage as shown in Equation (16). It can be shown that $log(\sum_{k \neq J+1} exp(\hat{v}_k)) - \hat{v}_{h_i} = log(1/\hat{P}_{h_i})$, so Equation (18) can be further simplified as:

$$log(1/P_{ih_i}) - \gamma_{0i} - \gamma_{1i} P_{h_i} rfr \tag{19}$$

4.1 Calculating New Equilibrium Following Policy Shifts

As a baseline case, one could assume that shifting rfr policy does not have any equilibrium effects on lifetime utilities, v and house values, P, i.e., $v_i(rfr) = v_i$ and $P_{h_i}(rfr) = P_{h_i}$. This would lead to an accurate representation of welfare effects only if the policy were applied to a very small number of households. Calculating such an

estimate would be straightforward with knowledge of γ_i, v, and P.

However, as the policy will affect all households, we may want to allow for equilibrium effects on both lifetime utilities, v, and house values, P. Doing this requires extending the framework of Bayer et al. (2016), which does not model equilibrium, to allow for equilibrium effects. This process necessitates making some straightforward parameterizations for the underlying flow utility function and transition matrices. Given those assumptions, we can model lifetime utilities, v, and house values, P, as functions of underlying primitives. Appendix B describes the calculation of new equilibria in detail.

5 Results

Our first stage model flexibly estimates household-by-neighborhood utility. Utility depends on neighborhood fixed effects and preference heterogeneity defined by interacting the household demographics in Section 2.2.1 with the neighborhood attributes in Section 2.2.2. We do not report full results in the paper because there are nearly 1,800 parameters. Appendix Table C.4 summarizes the interaction terms, $\hat{\Theta}$. They reveal strong and intuitive patterns of sorting and homophily. For example, college graduates sort toward neighborhoods with more college graduates, seniors toward neighborhoods with more seniors, and Black, Asian, and Hispanic households away from neighborhoods with larger White population shares.

As discussed above, for the second stage, we parameterize moving costs as: $MC_{ij} = \gamma_{0i} + \gamma_{1i} rfr \cdot P_j$. We allow for heterogeneity in both the baseline moving costs, γ_{0i} and in how realtor fees affect moving costs, γ_{1i}. We allow the baseline disutility of moving, γ_{0i}, to vary with neighborhood attributes, household demographics (age, marital status, income, education, and presence of school-aged children in the household) and a market fixed effect.[8] We also interact income with all neighborhood

[8] Allowing moving costs to vary with neighborhood characteristics recognizes that the physical cost of moving may vary with house size, age, and the user cost. Similarly, the psychic costs of

attributes, including realtor fees, so that γ_{1i} may vary by income.

Table 1: Second-Stage Coefficients

Realtor fee	-2.821^{***}
	(0.094)
Income × realtor fee	0.190^{***}
	(0.035)
Bachelor's degree	0.070^{***}
	(0.006)
Married	-0.320^{***}
	(0.006)
School-age child	-0.294^{***}
	(0.007)
Age: 30–35	-0.681^{***}
	(0.010)
Age: 35–40	-1.110^{***}
	(0.011)
Age: 40–45	-1.521^{***}
	(0.012)
Age: 45–50	-1.879^{***}
	(0.012)
Age: 50–55	-2.156^{***}
	(0.012)
Age: 55–60	-2.368^{***}
	(0.012)
Age: 60–65	-2.526^{***}
	(0.012)
Age: 65+	-2.815^{***}
	(0.010)
Mean disutility of moving (2026$)	169,700

Note: $N = 3{,}510{,}577$. The realtor fee is 6% of the household's home value (measured in $100,000s). The regression includes 149 market fixed effects, main effects for household demographics and neighborhood attributes, interactions between income and each neighborhood attribute, and an offset carrying the household's first-stage location-choice value. Standard errors in parentheses. Age enters as five-year bins with 18–30 as the reference category. The mean disutility of moving is computed as an average over the household-specific fitted utility gap between staying and moving, divided by the household's own income-specific fee sensitivity. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The estimates are shown in Table 1. We find that the disutility of moving

moving may vary with neighborhood demographics to the extent that residents are more or less embedded in neighborhoods with different demographic compositions. Alternatively, including neighborhood characteristics in the moving cost function can be interpreted as a control function for measurement error in the first-stage estimates for household-by-neighborhood lifetime utility. We show in Section 7 that the results of our counterfactual policy analyses are robust to either interpretation.

increases monotonically with age. As expected, we also see the disutility of moving is higher for married couples and for households with children. The realtor fee has a negative coefficient and a positive interaction with income, indicating that the marginal utility of additional income is lower for high income groups. Here income is measured in $100,000. When we use the marginal utility of income to monetize total moving costs, we get a mean close to $169,700 in 2026 dollars.

6 Implications for Sorting and Misallocation

6.1 Policy Simulations

To understand the impact of mobility frictions of household welfare and misallocation, we begin by examining the welfare effects of three counterfactual policies. Then, in Section 6.2, we consider optimal policy design under political feasibility constraints.

For the preliminary results that follow, we show the baseline case where we assume that shifting rfr policy does not have any equilibrium effects on lifetime utilities, v and house values, P, i.e., $v_i(rfr) = v_i$ and $P_{h_i}(rfr) = P_{h_i}$.

Our first counterfactual cuts the realtor fee rate from 6% to 3%. This scenario is motivated by the idea that buyers and sellers are now free to negotiate fees with realtors. The intent of the US Justice department case against NAR was that bargaining would cause the rate to decline. However this has not occurred. A more direct way to achieve the Justice department's goal would be to fix the rate at 3%.

Our second counterfactual sets a common flat fee across the US. By design, this eliminates the extreme variation in fees shown in Figure 1, raising fees in low-fee areas and reducing them in high-fee areas. This thought experiment is motivated by the idea that the tasks performed by realtors (e.g., pricing, marketing, closing) are similar across house values and geographies. We set the flat fee equal to the median realtor fee of $19,900.[9]

[9]Appendix Table C.1 reports a complementary version of this counterfactual that instead uses

Our third counterfactual strips realtor fees of the effects of the past 40 years of housing prices growth. Specifically, we keep the 6% rate but ask what would have happened had fees grown with consumer prices since 1986, instead of housing prices. This policy deflates overall fees as the U.S. has experienced real housing appreciation since 1986. It also affects spatial variation as there has been substantial spatial variation in real house price growth since 1986.[10] In this scenario, fees are deflated where real house prices actually rose, (e.g., San Francisco's fees are deflated by approximately half) and fees are inflated where real house prices actually fell (e.g., Worcester's fees are inflated by 6%).

For all three simulations, we calculate individual-specific changes in mobility probabilities and expected consumer surplus gains. Then we analyze variation within and between the 150 housing markets. We also calculate the change in total realtor commissions.

Table 2 shows the welfare affects on households and realtors, as well as the impact on within-market mobility. All three policies are welfare enhancing. Cutting fees by half has the largest welfare gain of $22.0 billion per year. It also has the largest effect on mobility, increasing it by 32%. Interestingly, equalizing realtor fees at the median yields very similar welfare gains, $21.4 billion, but with a lower effect on total mobility, and a much lower negative impact on realtor revenue.[11] Finally, we see that stripping realtor fees of the effects of appreciation of real housing prices would have smaller welfare and mobility gains. An interesting feature of this counterfactual is its ability to speak to Figure 2. The increase in realtor fees caused by housing price appreciation explains 14% of the reduction in homeowner mobility since 1986.

Table 3 uses two steps to decompose Table 2's changes in total annual welfare

the mean fee of $25,700.

[10] Our benchmark measures consumer prices with the CPI for all items less shelter. Appendix Table C.1 reports a more conservative variant that instead uses the housing CPI (the shelter index), under which fees track the cost of housing services. A further complementary exercise would shut down only the second effect, setting real house price growth equal to its spatial mean.

[11] Recognizing that reducing realtor commissions may be politically unpopular, we consider revenue neutral policies in Section 6.2.

Table 2: Main Counterfactual Results

	Halve fees	Equalize fees	Deflate fees
Mean CS change ($)	745	530	351
Change in realtor revenue ($M)	-18,660	-7,582	-5,122
Change in total annual welfare ($M)	22,035	21,386	14,060
% change in homeowner churn	32.3%	24.7%	16.0%
% of 1986–2018 homeowner migration decline offset	27.7%	21.2%	13.8%

Note: Dollar amounts here and throughout the paper are annual totals for the study-area population, computed with ACS household weights and expressed in constant 2026 dollars (CPI-U). Equalize fees sets every household's fee to the median current PUMA-level fee ($19,900); deflate fees scales each PUMA's fee by the ratio of non-housing CPI growth to local house-price growth since 1986. The last row multiplies the model's % change in churn by the within-market share of homeowner moves (79.6%, ACS estimation markets, 2013–2018) and by the CPS homeowner migration rate (4.3%, 2013–2018 average), then divides by the 1986–2018 decline in that series (8.3% to 4.3%). Between-market migration is held fixed; only within-market moves respond to the fee. If the counterfactual policies were to affect between-market moves as well, the bottom row would be larger.

into three distinct mechanisms. The first step asks what would happen to welfare if the policy changed but household behavior stayed fixed. This step yields the first mechanism, which is a *pure transfer*. At the household's original move probability, a lower fee simply shifts surplus from realtors to households. This is the largest object in every counterfactual but it has no effect on total welfare. The second step then asks what happens to household and realtor welfare when households re-optimize with respect to the new policy. The second mechanism is the re-optimization gain (relative to not responding to the new policy) to the household as they increase their utility by moving (or not) in response to the policy. Households facing lower fees increase welfare by gaining the surplus on newly attractive moves, while those facing higher fees mitigate their losses by reducing their probability of moving. The third mechanism is the corresponding effect for realtors. Each additional move induced by a fee cut increases realtor commissions and, conversely, each move that is deterred by a rising fee reduces commissions. Mechanisms two and three sum to the welfare row of Table 2. For example, if households did not respond to the Halve-Fees counterfactual policy, it would simply redistribute $33.7 billion from realtors to households. From this benchmark, the re-optimization of households will

increase their welfare by $7.0 billion and the welfare of realtors by $15.1 billion. The total welfare gain is therefore $22.0 billion (7.0+15.1) with gains for households of $40.7 billion (33.7+7.0) and losses for realtors of $18.7 billion (-33.7+15.1).

Table 3: Decomposition of the Change in Total Welfare ($M)

	Halve fees	Equalize fees	Deflate fees
Transfer at baseline behavior:			
fee payments saved by households	33,742	16,374	15,415
realtor revenue on baseline moves	-33,742	-16,374	-15,415
Welfare-relevant mechanisms:			
re-optimization surplus, fees fall	6,953	12,162	3,743
re-optimization surplus, fees rise	0	432	25
fees collected on induced moves	15,082	11,121	10,699
fees lost on deterred moves	0	-2,329	-406
Change in total annual welfare ($M)	22,035	21,386	14,060

Figure 5 summarizes spatial heterogeneity in how the counterfactual policies affect mobility and realtor commissions. It shows the percentage change in annual mobility across markets in the left column and the percentage change in realtor commissions across markets in the right column. Churn increases in all three cases. Some markets experience very large increases. In San Francisco, for example, halving fees or decoupling them from appreciation leads to a doubling of mobility and equalizing fees leads to a tripling.

While total realtor commissions fall in the aggregate for all three policies, there are many markets where commissions rise.[12] This occurs for two reasons. First, under fee equalization and decoupling, some areas experience a fee increase. Second, even in markets where the fee falls, realtor commissions can increase if the mobility response elasticity is greater than 1, i.e. if the market is to the right of the peak of a Laffer curve. In practice, we only see this in two markets, the San Francisco Bay Area and Santa Maria, where realtor commissions rise when fees are halved.

[12]Interestingly, we find that if fees were equalized at their mean, realtor commissions would increase on aggregate (Appendix Table C.1).

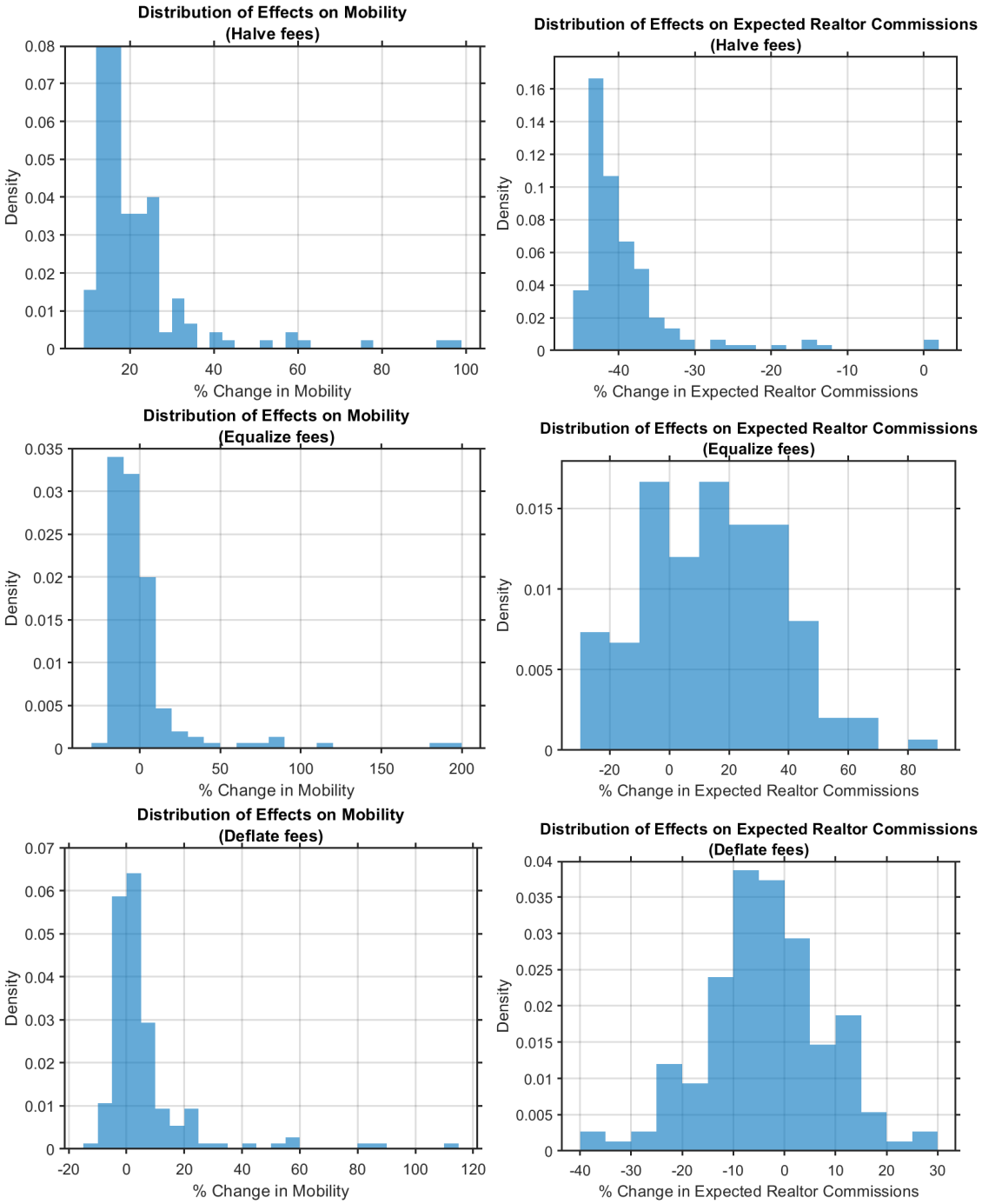
More economically relevant than the fact that two markets lie to the right of the Laffer-curve peak is the fact that there are many markets which lie to the left of that peak, but on a relatively flat portion of the curve. Their elasticities are only slightly below one so that halving the rate leads to relatively small reductions in commission rates and large welfare gains.

We can further illustrate how realtor fees lower household welfare (through misallocation and transfers) by calculating the consumer surplus from adjusting them. The heat map in Figure 6 shows the gain in expected annual consumer surplus. Lighter colors correspond to larger welfare gains. The gains are generally highest on the California coast, East Coast cities like New York and Boston, and relatively expensive housing markets in parts of Florida, Colorado, and Arizona. In those areas, the annual gains are often over $1,400 per household for all three policies. But for the average household, the gains range from $351 to $745. The consumer surplus gains tend to be larger in expensive coastal markets because the gains increase mechanically with housing value.[13] However, they are not perfectly correlated. This is because the gains also increase with diversity of housing choice. In large housing markets with more diverse neighborhoods to choose from, the expected gains from reducing the realtor fee are larger than in small housing markets where people choose from a smaller number of PUMAs that may be very similar.

Turning to realtor welfare, ex ante it is not obvious whether reducing the realtor fee rate will increase or decrease total commissions. The commission per sale is lower, but more people will move, which should increase total commissions. As noted above, commissions rise in a few extremely expensive markets with rate reductions because the increase in transaction volume offsets lower commission rates, but it is also notable that there are many coastal markets where realtor commission losses are small when rates are halved. In contrast, the markets that see realtor commission

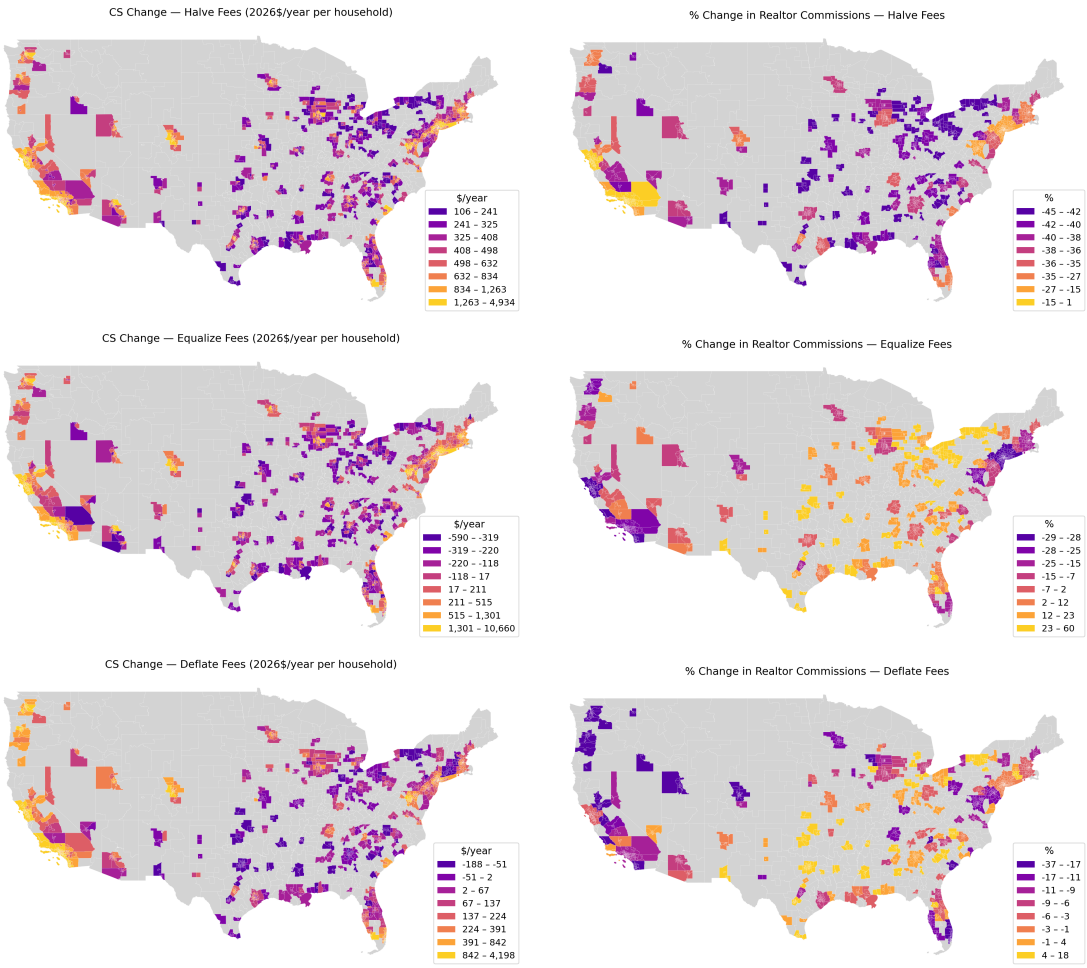
[13]Appendix Figure C.1 shows that the relationship between housing prices and mobility responses is strongly positive and that the relationship between baseline mobility and mobility responses is weak.

Figure 5: Heterogeneity in Counterfactual Effects on Mobility and Commissions



Notes: The left column shows the effect on mobility and the right column the effect on expected commissions, for halving fees (top row), equalizing fees (middle row), and deflating fees (bottom row). Each histogram weights the 150 markets equally. Under fee equalization commissions rise in 104 of 150 markets (the inexpensive ones, whose fees rise toward the median) while total revenue falls (Table 2). The declines concentrate in a few large, expensive markets that dominate the dollar totals.

Figure 6: Changes in Consumer Surplus and Realtor Commissions by PUMA

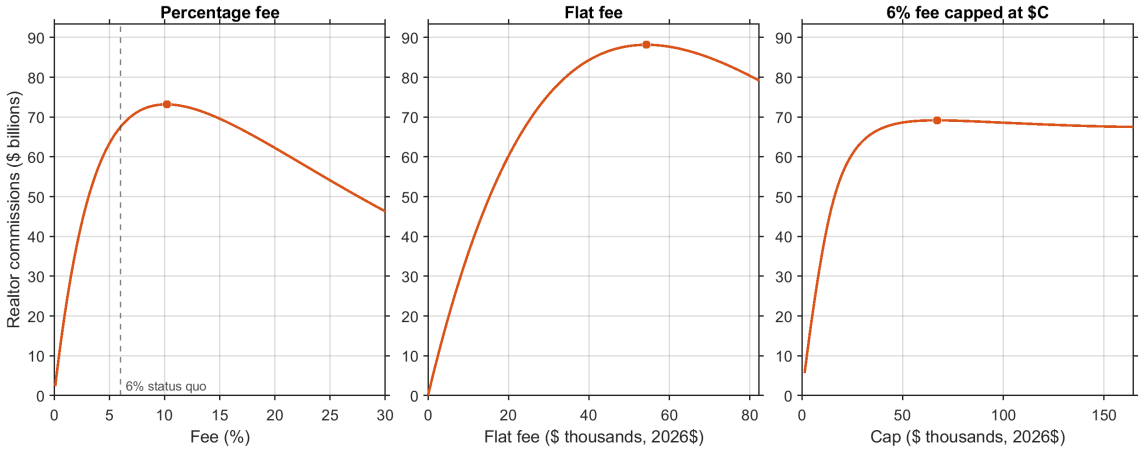


Notes: The left column shows the gain in expected annual consumer surplus by PUMA and the right column shows the percent change in realtor commissions by PUMA, for halving fees (top row), equalizing fees (middle row), and deflating fees (bottom row).

gains under equalization and deflation are the interior markets where fees increase.

Expanding on this idea, we calculate the Laffer curve for total realtor commissions for various realtor fee structures and parameters. Figure 7 shows this Laffer curve. The first panel shows the total commission as a function of a proportional rate. The optimal rate from a realtor perspective is 10.2%. However, the curve is relatively flat around this point and the existing rate of 6% yields 92.2% of the optimal commission amount. The second and third panels conduct an analogous exercise for two other fee structures: a national flat per-transaction fee and a cap on the 6% fee.

Figure 7: Realtor Commission Laffer Curves



Notes: Each panel plots aggregate realtor commissions as one fee instrument's parameter varies: the proportional rate (left), a national flat per-transaction fee (center), and a cap on the 6% fee (right). Dots mark each instrument's commission-maximizing point: a 10.2% rate, a $54,300 flat fee, and a $67,200 cap. The dashed line in the left panel marks the 6% status quo, which yields 92.2% of the commissions attainable at the 10.2% peak.

6.2 Policy Reform

The three counterfactual policies we simulated in the prior section each induced large welfare gains for households. However, they also generated substantial losses for realtors, which could limit their politically feasibility as policy reforms. With this in mind, we now consider optimal policy design subject to a distributional constraint. Specifically, we ask what fee structure would maximize household welfare subject to a constraint that total realtor commissions cannot fall?

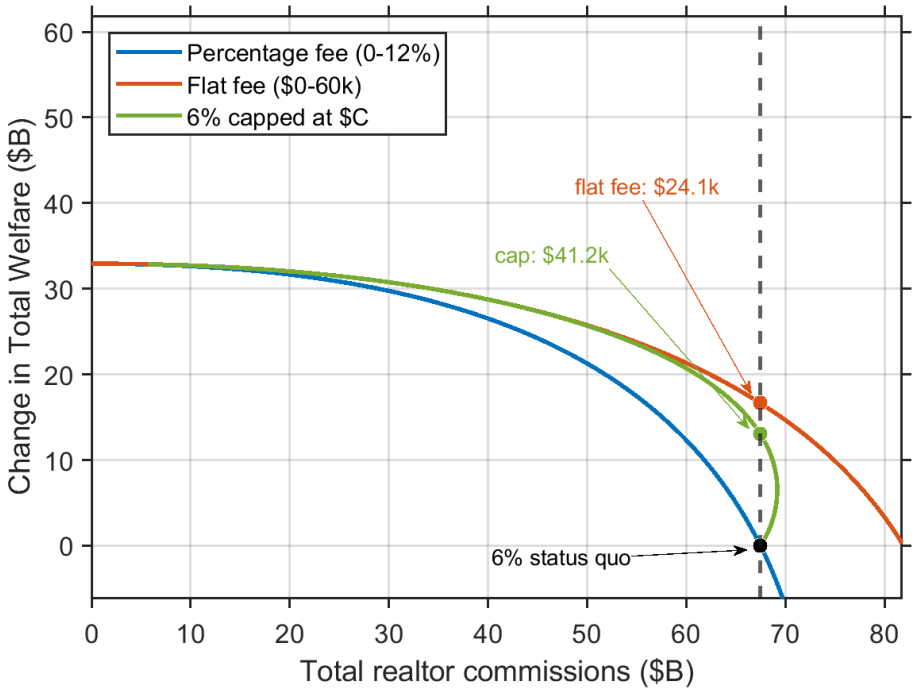
As an initial exercise, we consider three policies, each of which is governed by a single parameter: a proportional fee (the status quo is 6%), a flat per-transaction dollar fee, and a 6% fee capped at \$C per transaction. Figure 8 compares these policies. Each curve varies that policy's single parameter and plots the change in total welfare, relative to the 6% status quo, against total realtor revenue. The dashed vertical line marks status-quo revenue, so each curve's height at that line ranks the policies at equal realtor revenue; at that line the change in revenue is zero, so the height is exactly the policy's consumer-surplus gain. The optimal commission-neutral proportional fee is the 6% status quo; i.e., there is no Pareto improving policy when considering a proportional fee. The policy that keeps the 6% fee, but institutes a cap, finds an optimal cap of \$41,200 per transaction, which raises annual consumer surplus by \$13.0 billion. The constrained optimal flat fee is \$24,100 per transaction and raises annual consumer surplus by \$16.6 billion.[14]

While this exercise focuses on policies that don't reduce aggregate realtor commissions, it is worth noting that all three curves are quite steep around the status quo revenue line, but flat to the left of that line. The steepness around the status quo indicates that minor reductions in aggregate realtor commissions from the status quo can generate large gains in household welfare, thus generating gains in total welfare. However, continued cuts would only serve to transfer welfare from realtors to households.

By how much could welfare be increased by finer targeting? As a benchmark, we begin with the finest possible policy, which is to choose a separate realtor fee for each neighborhood. We again consider this policy subject to the constraint that realtor welfare doesn't fall. Thus, this policy is unconstrained with respect to the number of policy levers, but constrained with respect to realtor welfare. Specifically, this constrained first-best policy sets a dollar fee, D_p, in each of the 1,832 PUMAs

[14]We also considered two-parameter policies that combine a rate with a cap, or a fixed charge with a rate (a two-part tariff). In both cases, consumer welfare rises monotonically toward the flat-fee optimum of \$24,100, with zero cap or fixed charge. Appendix Figure C.3 shows how welfare varies across revenue-neutral cap/rate combinations.

Figure 8: Fee Structures Compared at Equal Realtor Revenue



Notes: Each curve varies one instrument's policy parameter—the proportional rate, the flat dollar fee, or the cap level—and plots the change in total welfare relative to the 6% status quo (the change in consumer surplus plus the change in realtor commissions) against total realtor commissions. The dashed vertical line is status-quo revenue ($67.5 billion); marked points are each instrument's revenue-neutral policy, whose height equals its consumer-surplus gain. All three curves climb toward the zero-fee corner at the upper left, where households gain $100.4 billion of surplus and realtors lose all $67.5 billion of revenue.

to maximize aggregate consumer surplus subject to not reducing aggregate realtor commissions:

$$\max_{D_1,\ldots,D_{1832}} \sum_i CS_i\big(D_{p(i)}\big) \qquad \text{s.t.} \qquad \sum_i D_{p(i)} \cdot P_i^{move}\big(D_{p(i)}\big) = R^0, \tag{20}$$

where $p(i)$ indexes household i's PUMA, CS_i is consumer surplus from Equation (11), P_i^{move} is the household's move probability from Equation (6), and R^0 is baseline revenue.

The problem of choosing the 1,832 fees can be written as a standard optimization

problem using a Lagrangian, λ, and an interior optimum every PUMA satisfies:

$$-\frac{\partial CS_p}{\partial D_p} = \lambda \frac{\partial R_p}{\partial D_p} \qquad \text{for all } p, \tag{21}$$

The Lagrange multiplier is the shadow cost of the realtor fee viewed as a per-transaction tax on moving. In other words, the last dollar of revenue raised through any PUMA's fee costs households λ dollars of surplus. We estimate $\lambda = 1.72$: the marginal revenue dollar costs households \$1.72, of which \$1 is the transfer itself and \$0.72 is deadweight loss. A lump-sum instrument, that ignored the realtor welfare constraint, would yield $\lambda = 1$. The annual welfare gain of this policy relative to the status quo is \$17.0 billion.

We now consider how much of the benchmark, first-best welfare gains could be obtained by a much simpler policy in which the planner may only choose a single national flat fee. This policy, by construction, will be weakly dominated by the first-best policy. In contrast, it will weakly dominate the status quo. However, choosing the national flat fee to maximize total household welfare can generate winners and losers among households. Therefore, we also consider policies that are constrained to both maintain total realtor welfare and not reduce household welfare relative to the status quo in any PUMA.

Table 4 reports how much of the first-best gains the simpler policies capture, with every row facing the same national revenue constraint. The first striking result, shown in row two, is that the national flat fee captures 98% of the first best welfare gains. However, it does so by raising fees on inexpensive homes, leaving 62.7% of households worse off relative to the status quo.

Rows three and four of Table 4 describe alternate policies designed to protect household welfare. These policies maintain the status quo 6% realtor fee rate, but cap fees in order to ensure that no households have reduced welfare. In row three, we allow PUMA-specific caps and, in row four, we only allow a single national cap. These policies only allow fees to fall, so they protect every household by construction.

Lowering fees in areas where the caps bind increases churn by just enough to leave aggregate realtor commissions unchanged.

As in the case of flat fee optimization, there is almost no difference between setting PUMA-specific policies and a national one. The second striking result in Table 4 is that a policy that is constrained to harm no households and has only a national cap, can obtain 76% of the first-best outcome. The λ column, which reflects sub-optimal distortions, shows the cost of the Pareto improving constraints. The first-best benchmark policy has a shadow value of 1.72. Restricting to a single flat fee has a minimal effect and increases it slightly to 1.74. Adding the additional Pareto constraint that no households are harmed roughly doubles λ to about 2.85. In summary, a national flat fee is nearly first-best but sharply regressive, leaving a majority of households paying more. A national cap on fees that otherwise maintains the 6% fee rate eliminates a quarter of the first-best gains to make no household worse off.[15]

Table 4: Share of the First-Best Welfare Gain Captured by Simple Policies

	Aggregate CS gain ($M)	% of first best	% losers	Marginal cost of revenue (λ)
PUMA-specific fees (first best)	17,026	100	58.8%	1.72
National flat fee ($24,100)	16,631	98	62.7%	1.74
PUMA-specific caps	13,561	80	0%	2.75
National cap ($41,200)	13,016	76	0%	2.85

Note: Aggregate gains are annual totals for the study-area population, computed with ACS household weights, in constant 2026 dollars. Every row holds total realtor revenue at its baseline level ($67.5 billion). The first best sets one fee in each of the 1,832 PUMAs; the cap policies set one cap per PUMA or one national cap. % losers: the share of households worse off than under the status quo. λ: the marginal consumer-surplus cost of the last revenue dollar under that instrument set.

There are two reasons why choosing a single national fee yields most of the welfare gains. First, when fees are allowed to vary across PUMAs, the degree of variation in the optimal PUMA-specific fees is relatively small. 99.4% of households

[15]These conclusions persist when we add flexibility to the second stage model that lets the marginal disutility of the realtor fee vary with all household demographics. The first-best gain is larger there, but the flat fee still captures 97% of it while making a majority of households worse off, and there is still minimal gain to allowing geographic granularity of fees or caps.

face first-best fees between $16,500 and $32,900, compared to the status quo in which the actual 6% fee generates a range from $3,800 to $174,200. Thus, the first best policy substantially compresses the fee distribution, raising fees for 58% of households while cutting them for 42%.[16] The second, related reason is that the PUMA-specific optimal fee function is relatively flat close to the PUMA-specific optimal fees, but steep further away. The local flatness explains why a single national fee captures most of the first-best gains. The non-local steepness explains why the first-best gains are so large relative to the status quo.

7 Robustness and Sensitivity Analyses

To assess the robustness of our results to modeling assumptions, we repeat the estimation and counterfactual simulations under two alternate specifications. Each specification is designed to address a potential concern about our identification strategy. We briefly summarize the results here, and report the counterfactual welfare measures in Appendix Table C.3.

First, we repeat estimation using an extended set of neighborhood amenities that includes six additional variables: school district quality level and growth, air-toxics cancer risk, transportation noise, walkability, and job access. Adding these amenities addresses the potential concern that our first-stage estimates for household-by-neighborhood utility values may embed measurement errors due to the practical need for parametric restrictions on the utility function. We find that adding the additional amenity variables hardly affects our main results. The signs and ordering of welfare measures across our counterfactuals are unchanged. The only notable difference is that the counterfactual welfare effects shrink by roughly ten percent (e.g. yielding a welfare gain of $19.7 billion rather than $22.0 billion when fees are halved).

[16] Appendix Figure C.2 provides a graphical comparison between the first-best and actual fee distributions.

The second sensitivity check is a control-function specification designed to address a potential concern with identifying the second-stage model parameters. In the second-stage, the attributes of a household's current neighborhood enter its moving cost, so a household in a desirable neighborhood is expensive to dislodge. Those same attributes could instead belong to the value of staying: what makes a neighborhood hard to leave is also what makes it good to live in, and the move/stay margin cannot separately identify these two effects. The control function imposes the second interpretation. In this interpretation, the second stage is effectively used to update the first-stage estimates of v_{ij}, thus correcting for any first-stage error/noise due to the first stage being misspecified or being estimated using a finite sample. Encouragingly, the counterfactual results are nearly indistinguishable from the baseline, so our policy conclusions do not depend on which interpretation we take.

8 Conclusion

We developed and estimated a dynamic model of residential sorting to examine how moving costs and realtor commissions affect household mobility, welfare, and the spatial allocation of households across neighborhoods *within* U.S. metropolitan areas. While local housing markets play a central role in determining access to economic opportunity and environmental quality, the institutional frictions embedded in those markets have received comparatively little attention in the residential sorting literature. We showed that realtor fees constitute a quantitatively important mobility friction that contributes to residential misallocation by substantially reducing how often households re-optimize housing consumption within their metropolitan housing markets.

These frictions have major implications for welfare and allocative efficiency. We found that reducing realtor commissions from 6 percent to 3 percent, to be more in line with international norms, would increase within-metro mobility by

approximately 32 percent and increase social welfare by roughly $22 billion per year. The largest opportunities for welfare gains exist in metropolitan areas where households face both high housing costs and diverse neighborhood choices. We also showed that modifying realtors' compensation structure could generate large efficiency gains without reducing their total compensation. For example, we showed that maintaining the prevailing 6% rate while imposing a national cap on fees of approximately $41,000 could generate $13 billion in annual welfare gains for homeowners without changing aggregate payments to real estate agents.

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Supplemental Appendix

A Data

A.1 Sample Selection

The estimation sample is constructed from the American Community Survey (ACS) microdata for the 2013–2018 period. We begin by linking each household to a metropolitan housing market using a series of geographic crosswalks that are associated with both the household's current Public Use Microdata Area (PUMA) and, for movers, the PUMA in which the household resided one year earlier (migPUMA). These metropolitan markets and how they are constructed are described in Section 2.

Our estimation sample is restricted to households that resided in one of the fifty U.S. states or the District of Columbia one year prior to the survey. We further restrict the sample to owner-occupied households, as these are the households facing Realtor fees upon moving. We focus on household heads to ensure a single observation per household, and include only adults aged 18 years or older. We also exclude household heads belonging to residual race and ethnicity categories with relatively few observations to ensure adequate sample sizes for demographic-specific estimates.

We next impose a series of geographic restrictions designed to ensure that both current and prior residential locations can be consistently assigned to metropolitan housing markets. First, we exclude households residing in PUMAs that cannot be assigned with at least 99 percent certainty (based on population shares) to a single metropolitan market. We then exclude destination PUMAs for which movers' origin geography cannot be reliably resolved into a market, in order to minimize potential measurement error due to incorrect market assignments. This is done because migration PUMAs are too geographically coarse to maintain meaningful variation in prices and other relevant variables. Finally, after removing those without

valid domestic migration histories, we restrict the sample to those whose previous residence can be confidently classified as originating within the same metropolitan housing market (within-market movers). We choose to focus on within-market moves because this type of housing churn driven by a desire to re-sort is more likely to be responsive to changes in Realtor fees. Cross-market moves, often driven by job changes, stage of life, etc., are systematically different from these and would contaminate our estimation as a result.

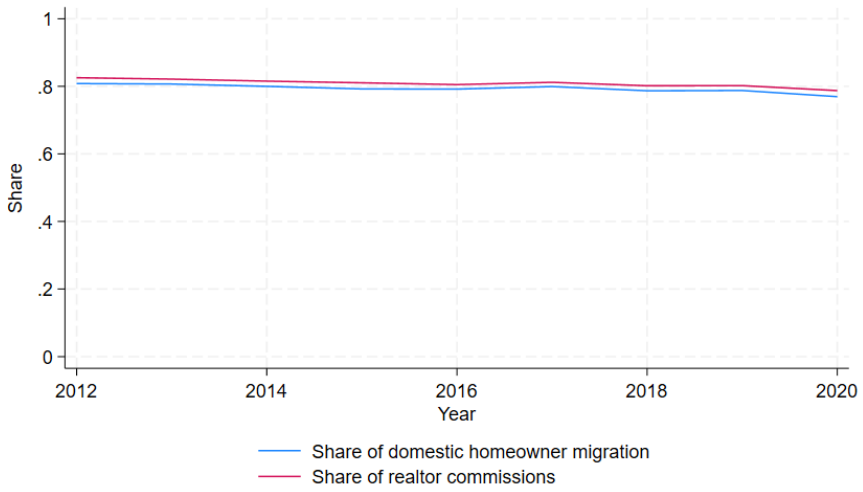
As a final sample restriction, we exclude households with annual income exceeding $1 million, as these observations are likely to be top-coded and behave in ways that are not representative of the general population. The result of these cuts leaves us with 3,510,577 observations in our final estimation sample. Table A.1 summarizes the sequence of sample restrictions and reports the number of observations remaining after each step.

Table A.1: Sample Construction

			Percent of All	
	Sample restriction	# Households	Households	Homeowners
(1)	All US households, 2013–2018	7,389,725	100.0%	100.0%
(2)	Lived in one of the 50 states/DC one year ago	7,360,480	99.6%	99.8%
(3)	Homeowner households	5,160,086	69.8%	99.8%
(4)	Head is an adult (18+)	5,159,384	69.8%	99.8%
(5)	Drop small race/ethnicity categories	5,054,256	68.4%	97.8%
(6)	Currently reside in a market (CSA or MSA)	3,767,587	51.0%	72.9%
(7)	PUMA's origin geography is resolvable	3,741,073	50.6%	72.4%
(8)	Movers: moved within the same market	3,696,591	50.0%	71.5%
(9)	Household income between $0 and $1,000,000	3,693,316	50.0%	71.4%
(10)	Market has more than 1 PUMA (estimation sample)	3,510,577	47.5%	67.9%

(1) Restricts the ACS sample to 2013–2018. (2) Drops households that lived abroad one year before the survey. (3) Drops renter households. (4) Drops households whose head is under 18. (5) Drops households whose head is not White (non-Hispanic), Black, Asian, or Hispanic. (6) Drops households whose PUMA is not at least 99% contained in a single market (CSA, or MSA not in a CSA). (7) Drops PUMAs whose own migration PUMA cannot identify the mover's origin market. (8) Drops movers whose origin market differs from their destination market. (9) Drops households with negative income or income above $1,000,000. (10) Drops households in markets with only one sampled PUMA, since the discrete-choice model needs multiple within-market alternatives.

Figure A.1: Within-Market Share of Homeowner Moves Over Time



Notes: The share of domestic homeowner migration that occurs within a market and the share of realtor commissions paid on within-market moves, by year.

A.2 Metropolitan Market Definitions and PUMAs

Our empirical analysis requires assigning each household to a metropolitan housing market that defines its relevant set of location alternatives. This market definition is designed to satisfy two objectives: first, markets should correspond to economically meaningful geographic areas within which households plausibly search for housing. Second, households must be assigned to markets with a high degree of confidence using only the geographic information available in the public-use ACS, thereby minimizing the possibility of erroneous market assignments.

We construct metropolitan markets by combining PUMAs, Metropolitan Statistical Areas (MSAs), and Combined Statistical Areas (CSAs). We first assign each PUMA to an MSA using a population crosswalk provided by IPUMS. Because PUMAs and MSAs are not defined on the same geographic boundaries, some PUMAs overlap multiple MSAs. To ensure that market assignments are effectively unique, we assign a PUMA to an MSA only when at least 99% of the PUMA's population resides within a single MSA. This assignment criterion is intended to minimize incorrect

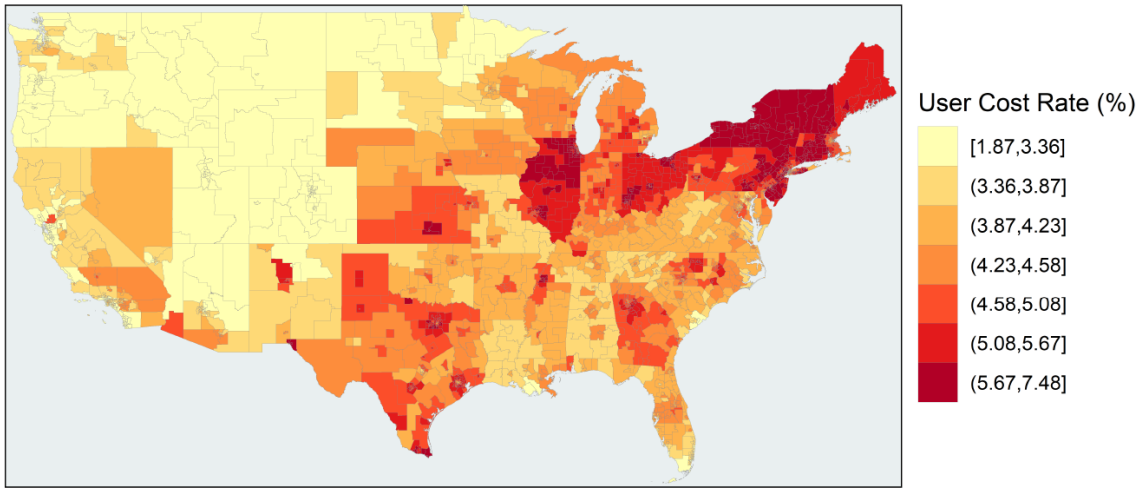
market assignments, as the public ACS only provides geographic identifiers down to the PUMA level, and households residing in PUMAs that span multiple MSAs cannot be assigned uniquely without introducing measurement error. As a result, PUMAs that do not satisfy this criterion are excluded from the analysis. The same process is done with migPUMAs to identify MSAs of origin.

We then aggregate MSAs into metropolitan markets using the Office of Management and Budget's CSA definitions. Specifically, if an MSA belongs to a CSA, the household's market is defined as the corresponding CSA. Otherwise, the MSA itself constitutes the market. This procedure produces a single set of metropolitan markets consisting of all CSAs together with standalone MSAs that are not members of a CSA. We apply the same procedure to the MSAs associated with migPUMAs in order to assign movers to their origin metropolitan markets.

The decision to define markets using the upper envelope of the CSA/MSA hierarchy reflects the objective of approximating the relevant housing market faced by households. CSAs encompass groups of closely connected metropolitan areas with substantial commuting and economic integration, providing broader and more realistic residential choice sets than individual MSAs while still maintaining well-defined geographic boundaries. Standalone MSAs remain separate markets when no broader CSA exists.

Finally, we exclude households residing outside metropolitan areas. These observations cannot be assigned to metropolitan housing markets under our definition and therefore lack well-defined residential choice sets consistent with our empirical framework.

Figure A.2: User Cost Rates by PUMA in 2016



B Equilibrium Solving Algorithm

Agents maximize the discounted sum of expected flow utilities, u and the only time varying state variables are h_{it} and ϵ_{ijt}. All other agent and neighborhood characteristics are time invariant as we assume that neighborhoods and individuals are in a long-run steady-state equilibrium similar to Davis et al. (2021). h_{it} transitions deterministically and endogenously as, by construction, $h_{it+1} = d_{it}$. As discussed above, ϵ_{ijt} transitions stochastically and exogenously and is distributed i.i.d. (over i, j and t) according to Type 1 Extreme Value. This satisfies the Conditional Independence assumption of Rust (1987) as ϵ_{ijt} has no predictive power regarding either ϵ_{ijt+1} or h_{it+1}, conditional on d_{it}.

Mirroring the notation for value functions, let

$$\tilde{U}_{ijt} = u_{ij} + \epsilon_{ijt} \tag{B.1}$$

and let

$$U_{ijt} = u_{ij} - MC_{ih_{it}} 1_{[j \neq J+1]} + \epsilon_{ijt}. \tag{B.2}$$

The choice specific value function, V_{ijt} is given by:

$$V_{ijt} = u_{ij} - (\gamma_0 + \gamma_1 P_{h_{it}} rfr) 1_{[j \neq J+1]} + \epsilon_{ijt} + \beta E_\epsilon [max\{V_{ikt+1}\} | d_{it} = j] \tag{B.3}$$

where

$$E_\epsilon [max\{V_{ikt+1}\} | d_{it} = j] = log\Big(exp(v_{ij}) + \sum_{k \neq J+1} exp(v_{ik} - (\gamma_0 + \gamma_1 P_j rfr))\Big) \tag{B.4}$$

and the net-of-moving-costs-and-shock choice specific value function, v_{ij} is given by:

$$v_{ij} = u_{ij} + \beta log\Big(exp(v_{ij}) + \sum_{k \neq J+1} exp(v_{ik} - (\gamma_0 + \gamma_1 P_j rfr))\Big) \tag{B.5}$$

Given the model assumptions, v_{ij} is not a function of time t and does not depend on current time-varying states ϵ_{it} and h_{it}. State transitions are straightforward to incorporate as h_{it} transitions deterministically and taking the expectation over ϵ gives the familiar log-sum closed-form solution.

For a given value of u_{ij} $\forall j$, P_j $\forall j$, and rfr, it is straightforward to solve for v_{ij} $\forall j$ via fixed-point iteration. A related result is that given estimates of v_{ij}, γ_{0i}, and γ_{1i}, it is trivial to solve for u_{ij} as:

$$u_{ij} = v_{ij} - \beta log\Big(exp(v_{ij}) + \sum_{k \neq J+1} exp(v_{ik} - (\gamma_0 + \gamma_1 P_j rfr))\Big) \tag{B.6}$$

Counterfactual v_{ij} for alternative rfr (holding fixed house values) are therefore easy to calculate given estimates of u_{ij}, P_j, and counterfactual policies, rfr', using Equation (B.5).

To further consider how rfr may affect house values, P, additionally partially parameterize u_{ij}

$$u_{ij} = f_i(amenity_j, \xi_j) - \alpha_{uc} uc_j \tag{B.7}$$

Recall that $uc_j = P_j ucr_j$. We have an estimate of α_{uc} as, by construction, $\alpha_{uc} = \gamma_1$. As we have estimates of u_{ij} and α_{uc}, we can recover as estimate of $f_i(amenity_j, \xi_j)$ using $f_i(amenity_j, \xi_j) = u_{ij} - \alpha_{uc} uc_j$.

Therefore, given any value of P_j, we can construct a $u_{ij}(P_j)$ using the fixed and known terms: α_{uc}, $f_i(amenity_j, \xi_j)$, and ucr_j. Holding fixed γ_0, γ_1, α_{uc}, $f_i(amenity_j, \xi_j)$, and ucr_j, then for a given set of P_j, we can solve for v_{ij}, where P affects v_{ij} through the moving cost and user cost terms.

Additionally using a distribution for h_i, we can then calculate the demand for each neighborhood, j:

$$Demand_j = \sum_i \Bigg(\frac{exp(v_{ij} - (\gamma_0 + \gamma_1 P_j rfr))}{exp(v_{h_i}) + \sum_{k \neq J+1} exp(v_k - (\gamma_0 + \gamma_1 P_j rfr))} + \frac{exp(v_{ij})}{exp(v_{h_i}) + \sum_{k \neq J+1} exp(v_k - (\gamma_0 + \gamma_1 P_j rfr))} 1_{[j=h_i]} \Bigg) \quad \text{(B.8)}$$

In other words, for any combination of counterfactual policy, rfr, housing values, P_j, and distribution of prior locations, $F(h_i)$, we can calculate demand and compare it with supply.

Then for a given rfr, we can solve for a vector of housing values, P_j that clear the market, given v_j. And, we can solve for a vector of value functions, v_j, that clear the market, given housing values, P_j. The key is then to iterate on these two problems.

We let supply be given by $Supply_j$, which may be fixed or parameterized using existing estimated supply elasticities, such as the $Supply_j = Supply(P_j)$ mapping taken from Baum-Snow and Han (2024) which is the strategy taken in Davis et al. (2021).

A summary of this of the equilibrium-solution algorithm is:

Step 0 Fix γ_0, γ_1, α_{uc}, $f_i(amenity_j, \xi_j)$, and ucr_j and some initial distribution of location choices, $F(h_i)$ at their estimates and observed levels. Initialize guesses of P_j, presumably at observed levels.

Step 1 For parameters fixed in 0 and current values of P_j, calculate u_{ij} using Equation (B.7). Then iterate on Equation (B.5) until convergence to solve for v_j.

Step 2 For parameters fixed in 0 and current values of v_j, use Equation (B.8) to create overall demand for each neighborhood, $Demand_j \equiv \sum_k P_j(d_{t-1} = k) f(d_{t-1} = k)$. Use the contraction mapping described in Berry et al. (1995) to update P_j (and therefore $Demand_j$) to make $Demand_j = Supply_j$.

Step 3 Iterate on 1 and 2 until convergence in v_j and P_j.

Each Step 1 and Step 2 are guaranteed to converge.

C Additional Tables and Figures

Table C.1: Counterfactual Results: Additional Policies

		Equalize		Deflate	
	Halve fees	median fee	mean fee	non-housing CPI	housing CPI
Mean CS change ($)	745	530	222	351	152
Change in realtor revenue ($M)	-18,660	-7,582	2,532	-5,122	-94
Change in total annual welfare ($M)	22,035	21,386	14,643	14,060	8,201
% change in homeowner churn	32.3%	24.7%	11.3%	16.0%	7.4%
% of 1986–2018 homeowner migration decline offset	27.7%	21.2%	9.7%	13.8%	6.3%

Note: Expands Table 2 with the two policy variants not featured in the text. The equalize columns set every household's fee to the median ($19,900) or mean ($25,700) current PUMA-level fee, in 2026 dollars. The deflate columns scale each PUMA's fee by the ratio of CPI growth to local house-price growth since 1986, using either the non-housing CPI (all items less shelter) or the housing CPI (the shelter index). The last row is constructed as described in the note to Table 2.

Table C.2: Decomposing Welfare Changes ($M): Additional Policies

		Equalize		Deflate	
	Halve fees	median fee	mean fee	non-housing CPI	housing CPI
Transfer at baseline behavior:					
fee payments saved by households	33,742	16,374	1,237	15,415	5,907
realtor revenue on baseline moves	-33,742	-16,374	-1,237	-15,415	-5,907
Welfare-relevant mechanisms:					
re-optimization surplus, fees fall	6,953	12,162	9,348	3,743	2,198
re-optimization surplus, fees rise	0	432	1,527	25	189
fees collected on induced moves	15,082	11,121	10,476	10,699	7,567
fees lost on deterred moves	0	-2,329	-6,707	-406	-1,754
Change in total annual welfare ($M)	22,035	21,386	14,643	14,060	8,201

Note: Expands Table 3 with the two policy variants not featured in the text; the policy columns are defined in the note to Table C.1, and the decomposition is constructed as described in the note to Table 3.

Table C.3: Counterfactual Effects Under Alternative Specifications

	Halve fees	Equalize fees	Deflate fees
Panel A: Change in total annual welfare ($M)			
Main specification	22,035	21,386	14,060
Extended amenities	19,718	18,813	12,534
Control function	22,151	21,526	14,135
Panel B: % change in homeowner churn			
Main specification	32.3%	24.7%	16.0%
Extended amenities	28.8%	20.8%	14.2%
Control function	32.4%	25.1%	16.1%

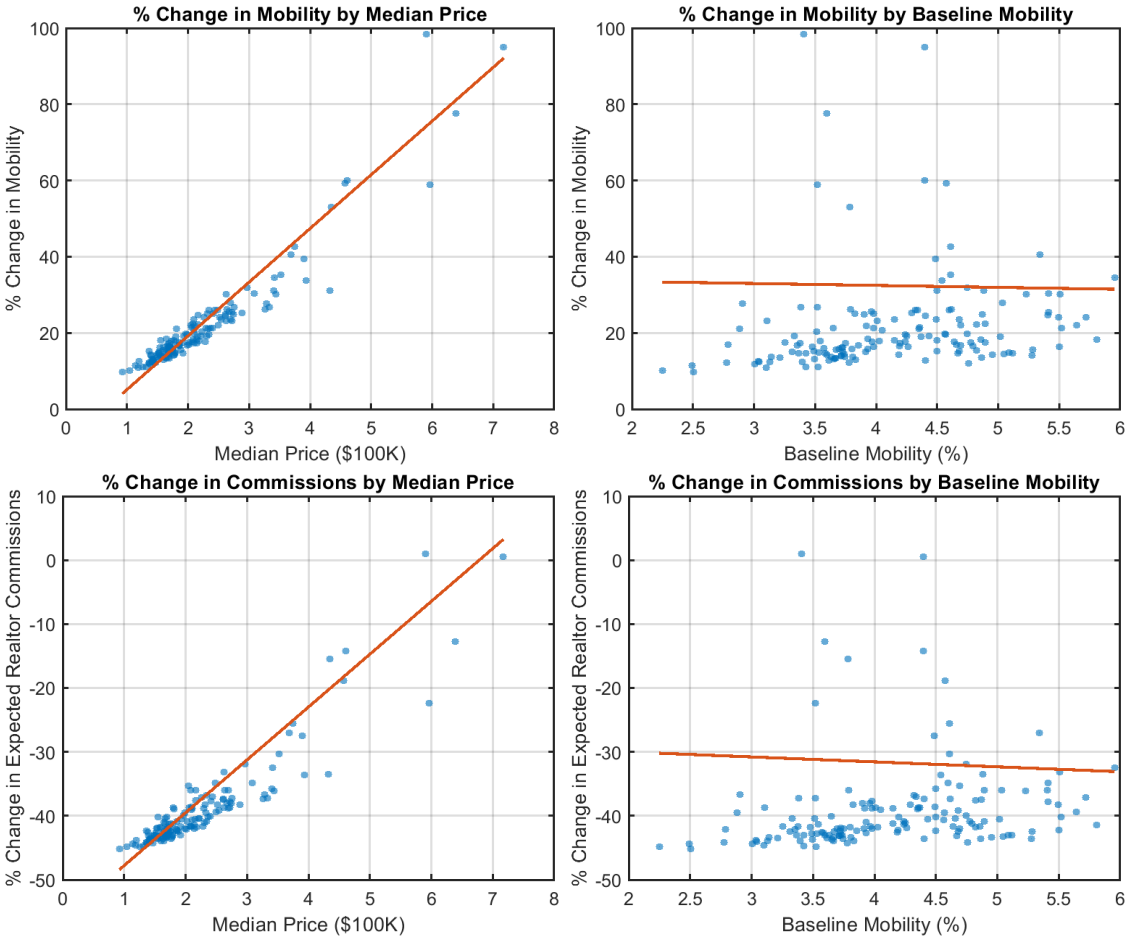
Note: For each alternate specification shown in the table, we re-estimate both stages of the model and repeat each counterfactual. "Control function" reassigns the origin-amenity coefficients from the moving cost to the flow utility of staying. "Extended amenities" adds six neighborhood attributes (school quality level and growth, air-toxics cancer risk, transportation noise, walkability, and job access). School quality is measured by the level and the growth of standardized test scores in the PUMA's school districts, from the Stanford Education Data Archive (SEDA version 5.0, restricted to the 2013–2018 school years) and assigned to neighborhoods through the National Center for Education Statistics school-district boundary files. Air-toxics cancer risk is the lifetime cancer risk from airborne toxics in the EPA's 2014 National Air Toxics Assessment. Transportation noise is the share of the population exposed to average transportation noise of 50 decibels or more, built from the U.S. Department of Transportation's National Transportation Noise Map. Walkability is the National Walkability Index from the EPA's Smart Location Database, and job access is the number of jobs reachable within a 45-minute automobile commute, from the same source. Each measure is aggregated from Census tracts (or block groups for the Smart Location Database) to PUMAs using population weights and standardized. This specification also drops two Opportunity Atlas measures (marriage and employment at age 29) to reduce multicollinearity in the amenity controls.

Table C.4: First-Stage Interaction Coefficients $\hat{\Theta}$

	Coef.	(s.e.)		Coef.	(s.e.)
College share × income	0.908***	(0.066)	User cost × married	0.454**	(0.206)
College share × senior	0.351**	(0.171)	User cost × school-age child	0.601**	(0.235)
College share × bachelor's	4.027***	(0.130)	User cost × Black	−2.671***	(0.544)
College share × married	−0.491***	(0.122)	User cost × Asian	0.575**	(0.281)
College share × school-age child	−0.739***	(0.134)	User cost × Hispanic	−2.665***	(0.368)
College share × Black	1.745***	(0.259)	PM2.5 × income	−0.005	(0.007)
College share × Asian	3.789***	(0.208)	PM2.5 × senior	−0.048***	(0.016)
College share × Hispanic	−2.062***	(0.212)	PM2.5 × bachelor's	0.013	(0.012)
Mean income × income	0.408***	(0.043)	PM2.5 × married	0.005	(0.012)
Mean income × senior	0.348**	(0.136)	PM2.5 × school-age child	−0.001	(0.013)
Mean income × bachelor's	0.120	(0.108)	PM2.5 × Black	−0.065**	(0.026)
Mean income × married	−0.057	(0.098)	PM2.5 × Asian	0.082***	(0.020)
Mean income × school-age child	0.105	(0.110)	PM2.5 × Hispanic	0.024	(0.017)
Mean income × Black	−1.036***	(0.227)	OA incarceration × income	−0.145	(0.428)
Mean income × Asian	−0.729***	(0.143)	OA incarceration × senior	0.838	(0.900)
Mean income × Hispanic	1.332***	(0.171)	OA incarceration × bachelor's	−0.041	(0.654)
White share × income	−0.147***	(0.033)	OA incarceration × married	−0.020	(0.637)
White share × senior	−0.849***	(0.071)	OA incarceration × school-age child	−0.543	(0.676)
White share × bachelor's	−0.589***	(0.055)	OA incarceration × Black	0.603	(1.045)
White share × married	0.235***	(0.052)	OA incarceration × Asian	1.857*	(1.077)
White share × school-age child	0.118**	(0.058)	OA incarceration × Hispanic	2.102*	(1.165)
White share × Black	−6.040***	(0.090)	OA income at 29 × income	−0.561***	(0.171)
White share × Asian	−5.033***	(0.092)	OA income at 29 × senior	0.589	(0.435)
White share × Hispanic	−4.242***	(0.083)	OA income at 29 × bachelor's	−0.802***	(0.301)
Mean age × income	0.001	(0.015)	OA income at 29 × married	1.199***	(0.296)
Mean age × senior	−0.010	(0.040)	OA income at 29 × school-age child	0.674**	(0.319)
Mean age × bachelor's	−0.077**	(0.030)	OA income at 29 × Black	5.601***	(0.654)
Mean age × married	0.085***	(0.028)	OA income at 29 × Asian	6.775***	(0.463)
Mean age × school-age child	0.171***	(0.033)	OA income at 29 × Hispanic	8.404***	(0.573)
Mean age × Black	0.159***	(0.059)	OA married at 29 × income	−0.013	(0.080)
Mean age × Asian	0.199***	(0.049)	OA married at 29 × senior	0.348*	(0.206)
Mean age × Hispanic	−0.446***	(0.046)	OA married at 29 × bachelor's	−0.379***	(0.138)
Senior share × income	−0.080	(0.213)	OA married at 29 × married	0.427***	(0.135)
Senior share × senior	5.149***	(0.530)	OA married at 29 × school-age child	0.525***	(0.153)
Senior share × bachelor's	0.811*	(0.416)	OA married at 29 × Black	−2.128***	(0.341)
Senior share × married	−0.836**	(0.389)	OA married at 29 × Asian	−0.101	(0.225)
Senior share × school-age child	−2.730***	(0.454)	OA married at 29 × Hispanic	−1.208***	(0.299)
Senior share × Black	−0.514	(0.828)	OA employed at 29 × income	0.083	(0.138)
Senior share × Asian	−3.498***	(0.703)	OA employed at 29 × senior	−0.282	(0.323)
Senior share × Hispanic	4.733***	(0.625)	OA employed at 29 × bachelor's	0.256	(0.225)
Mean bedrooms × income	−0.111***	(0.019)	OA employed at 29 × married	−1.027***	(0.223)
Mean bedrooms × senior	−0.172***	(0.054)	OA employed at 29 × school-age child	−0.441*	(0.234)
Mean bedrooms × bachelor's	−0.215***	(0.041)	OA employed at 29 × Black	−0.213	(0.454)
Mean bedrooms × married	0.516***	(0.038)	OA employed at 29 × Asian	−0.322	(0.362)
Mean bedrooms × school-age child	0.553***	(0.044)	OA employed at 29 × Hispanic	−0.419	(0.406)
Mean bedrooms × Black	0.546***	(0.082)	OA teen birth × income	−0.803***	(0.103)
Mean bedrooms × Asian	0.257***	(0.061)	OA teen birth × senior	−0.180	(0.224)
Mean bedrooms × Hispanic	0.161***	(0.061)	OA teen birth × bachelor's	−0.711***	(0.159)
Median house age × income	−0.025***	(0.006)	OA teen birth × married	0.234	(0.154)
Median house age × senior	0.112***	(0.016)	OA teen birth × school-age child	0.462***	(0.164)
Median house age × bachelor's	−0.088***	(0.012)	OA teen birth × Black	1.506***	(0.277)
Median house age × married	0.072***	(0.011)	OA teen birth × Asian	−1.794***	(0.286)
Median house age × school-age child	0.033**	(0.013)	OA teen birth × Hispanic	0.504*	(0.266)
Median house age × Black	0.088***	(0.023)	OA incarceration (level)	−0.615	(1.048)
Median house age × Asian	0.091***	(0.020)	OA income at 29 (level)	−6.172***	(0.492)
Median house age × Hispanic	−0.061***	(0.019)	OA married at 29 (level)	−0.044	(0.237)
User cost × income	−0.003	(0.081)	OA employed at 29 (level)	0.906**	(0.374)
User cost × senior	0.452	(0.278)	OA teen birth (level)	−0.182	(0.268)
User cost × bachelor's	−0.402*	(0.229)			

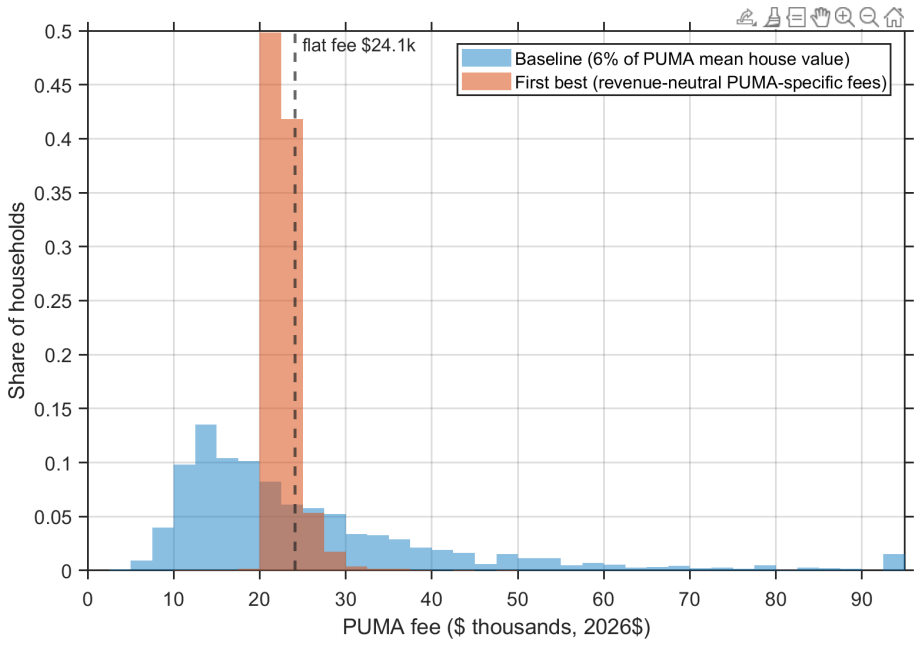
Note: Estimates from the first-stage conditional logit over within-market alternatives. The table reports interactions between PUMA attributes and household attributes. The model also includes 1,682 alternative-specific constants, which are not reported. Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure C.1: Effects of Halving Fees, by Baseline Home Value and Baseline Mobility



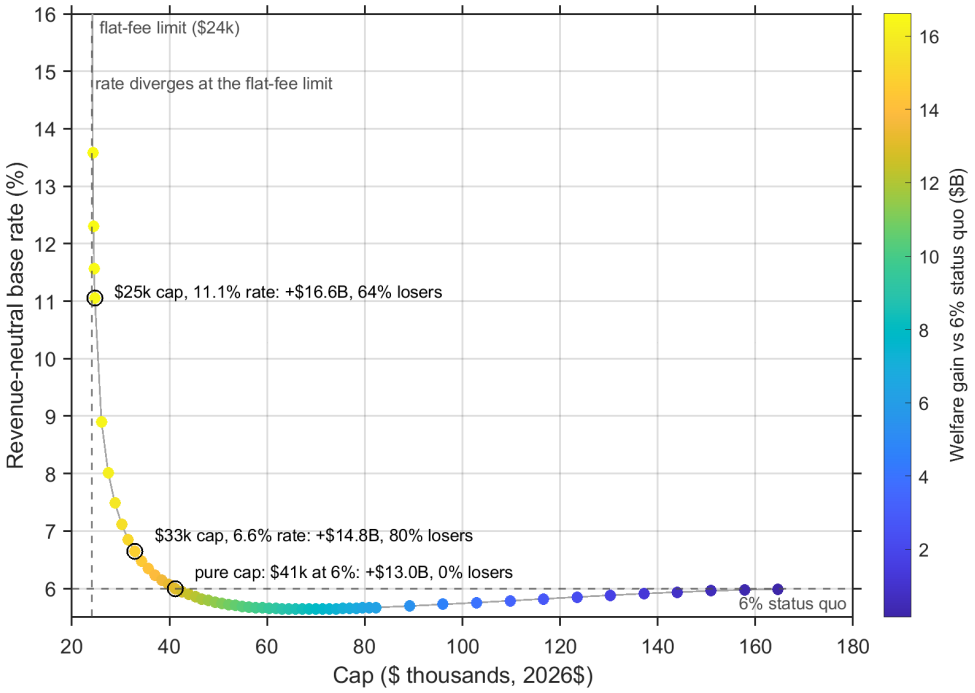
Notes: Each marker is one of the 150 estimation markets. Both rows show effects of halving fees: the change in mobility (top row) and the change in expected realtor commissions (bottom row), plotted against the market's median home value (left column) and its baseline mobility rate (right column). Fit lines are population-weighted; markers are unweighted.

Figure C.2: PUMA Fees at Baseline and Under the First Best



Notes: Household-weighted distributions over the 1,832 estimation PUMAs in $2,500 bins, in constant 2026 dollars; the last bin collects the 1.3% of households whose baseline fee exceeds $95,000. The dashed line is the revenue-neutral national flat fee of $24,100. Under the first best 99.4% of households face a fee between $16,500 and $32,900.

Figure C.3: Revenue-Neutral Cap/Rate Schedules



Notes: Each point on the curve pairs a cap with the base rate that maintains status-quo realtor revenue, so moving along the curve changes only the fee structure. The welfare gain over the status quo is denoted by differently colored dots along the curve. Every point holds realtor commissions at $67.5 billion, so the welfare gain equals the consumer-surplus gain. Caps and gains are in constant 2026 dollars. The curve has a vertical asymptote at the revenue-neutral flat fee. This is because every household's fee is at most the cap, so a cap below $24,100 cannot maintain baseline revenue at any rate, so the revenue-neutral rate diverges as the cap approaches the limit.